

Controlling Chaos: The OGY Method, Its Use in Mechanics, and an Alternative Unified Framework for Control of Non-regular Dynamics

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Abstract In this chapter we review the development of the control of chaos theory subsequent to the seminal paper by Ott, Grebogi and Yorke in 1990. After summarizing the main characteristics of the OGY method, we analyze and discuss various applications in several fields of mechanics. We then illustrate the main aspects of an alternative control method which aims at controlling the overall system dynamics instead of stabilizing a single periodic orbit, as the OGY method does. The two methods are both based on the modern idea of exploiting the chaotic properties of systems, instead of simply eliminating chaos.

This paper is one of a collection written in honour of Celso Grebogi, on the occasion of his 60th birthday. So we have thought it appropriate to start with short personal reminiscences by two of the present authors.

Some personal remarks by Michael Thompson.

Thinking back to my first encounters with chaos in the early 1980's, I remember that the work of "Grebogi, Ott and Yorke" soon became indelibly etched on my mind. Interested by some unexpected subharmonic resonances that had been observed when oil tankers were moored to articulated off-shore towers, I had modelled the system as a driven impact oscillator. This exhibited a lot of fascinating nonlinear phenomena, including what David Rand (at Warwick University) told me were chaotic motions: about which I had to do some very rapid reading! My first publication about the chaos in this impacting system [1] appeared in 1982 which was, interestingly, the year of Celso's first major paper on chaos [2]. The latter, authored by Grebogi, Ott and Yorke, was a seminal paper dealing with the bifurcational crises of chaotic attractors: to date it has attracted 483 citations. It became one of my standard references in, for example, my paper on the design of compliant off-shore structures [3].

Having linked up with Bruce Stewart at the Brookhaven National Laboratory, New York, our book [4] entitled *Nonlinear Dynamics and Chaos* (Thompson and

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Stewart, published by John Wiley) appeared in 1986. Key references at the end of this book included Celso's second paper on crises [5] (838 cites)), and his work on quasi-periodicity of order three [6] (76 cites).

I next turned my attention to the chaotic phenomena that triggered the escape from a potential well, publishing in 1989 a paper in the *Proceedings of the Royal Society* [7] which itself has attracted 120 citations. In this work I found it necessary to reference six papers by Celso including [5] described above. A paper of particular significance to me, by McDonald et al. [8] (351 cites), was on fractal basin boundaries which I warned could seriously decrease the integrity of engineering systems.

From this spectacular start, Celso's career has gone from strength to strength, and he is now on the distinguished ISI List of Highly Cited Researchers. Meanwhile the writers of the present paper note with special pleasure and satisfaction that the paper entitled *Controlling Chaos* [9] published by Ott, Grebogi and Yorke in 1990 has been selected by the American Physical Society as one of the milestone pieces of research from the last 50 years. This puts Celso (alongside many Nobel Prize winners) in the list of milestone contributors which has been created by the society to celebrate the 50 years of *Physical Review Letters*. Not surprisingly, this paper of 1990 is Celso's most highly referenced work, having been cited 2,711 times.

After seeing his early papers in print, I soon enjoyed meeting Celso at many international conferences throughout the world: and now, I am delighted and honoured to be getting to know Celso and his family very well indeed because we both hold Sixth Century Chairs at the University of Aberdeen (his full-time, mine part-time).

Some personal remarks by Giuseppe Rega.

I first met Celso Grebogi at the IUTAM Symposium on *Nonlinearity and Chaos in Engineering Dynamics* organized by Michael Thompson at University College London, 1993. Within a top level specialty meeting of scientists from engineering mechanics working on nonlinear dynamics and their applications, Celso was invited, from a different environment, to mark a conscious and widespread transition from the traditional asymptotic/numeric treatment of nonlinear dynamics problems followed within the community of mechanicians, to a more modern and integrated approach relying additionally on geometric and experimental methods. Celso fully succeeded in his own task, by providing the attendees with a clear feeling of the importance of a dynamical systems approach to nonlinear/chaotic dynamics in engineering and by introducing the newly developed concept and technique of chaos control. Since then Celso has become for the engineering community the most appropriate and acclaimed scientist for knowledge transfer from Physics as well as for cross-fertilization between Physics and Engineering.

Within his extraordinary number of achievements on cutting edge issues in dynamical systems theory and applications and assuming an engineering mechanics scholar's viewpoint, I just mention two items. (i) All dynamical systems scientists from engineering have become familiar with concepts like sudden change bifurcations in chaotic attractors and basin boundary metamorphoses by reading Celso and co-authors' papers; (ii) the OGY method paved the way for a modern and

comprehensive notion of control of chaos, including its suppression, enhancement or use, which is of major importance for practical applications.

In the last decade, Celso's interests meaningfully spread to include problems of dissipative dynamics typically occurring in engineering applications. His former interests towards applications of chaos and control to, e.g., plasma and laser systems or communication, evolved to include such typical systems and problems in mechanics as pendula, ship capsizing, impact and dry friction oscillators, fluidized beds, hydrodynamical flows, suspension bridge models, and spacecraft steering.

Within the scientifically fertile and prestigious Maryland group, Celso gave fundamental contribution as regards both the scientific aspects and the worldwide promotion of knowledge on chaotic dynamics and their applications. His broadmindedness and capability in cross-disciplinary relationships allowed him to play a role of most visible and internationally recognized scientist by neighbouring scientific communities dealing with complexity. A capability to lead and motivate research groups and disseminate knowledge to companion communities and wider non-specialists' audiences that were fully confirmed when he moved to the University of Sao Paulo and, more recently, to the University of Aberdeen.

1 Controlling Chaos: A Hot Topic at the Change of the Millennium

Among various other research topics which attracted the attention of researchers in the 20 years across the change of the Millennium, control of nonlinear dynamics and chaos is certainly one of the most important and well identified, developed at length from a few seminal ideas and methods, and yet far from being exhausted.

The quality of papers published on this topic cannot be judged by numbers or quantitative indicators, and indeed we found, as in other research fields, both very good – seminal, indeed – and poor works. However, numbers provide an understanding on the “impact” of this discipline on the advancement of science, engineering, practice and, more generally, of knowledge. Thus, in the certainly incomplete list of references we quote:

- 8 books [10–17] by the most important publishers;
- 9 journal special issues [18–26] of the most renowned scientific journals;
- 13 review or survey papers [27–39] in leading international scientific journals.

Even more impressive is the number of articles and conference papers published on the subject. According to “Scopus” (www.scopus.com), looking for “control of chaos” in the title, abstract or keywords, on 3 February 2009 we got 7,526 papers, with a growing rate of papers published per year. This extends the data of Fradkov et al. [33] who refer to 2,700 papers published in peer-reviewed journals up to the year 2000, with more than a half in 1997–2000.

One interesting aspect of the “control of chaos” discipline is the cross-disciplinarity, which is a consequence of the cross-disciplinarity of its parent, the “chaos theory.” Thus, we have papers from mathematics, physics, chemistry,

biology, medicine, economics, and various fields of engineering (see Sect. 3 for more details related to mechanical engineering and structural dynamics).

The starting point of the theory and applications of “chaos control” is unanimously considered to be the work of Ott, Grebogi and Yorke [9]. Actually, other works aimed at “suppressing chaos” in some systems appeared at about the same time or earlier (e.g., [40, 41]), but the seminal idea of exploiting the chaotic behaviour of systems in order to control their dynamics was first undoubtedly presented in the OGY paper. It represents a major improvement with respect to other points of view on the matter, independently pursued within different scientific communities (of mathematicians, physicists and engineers): in fact, it marks the fundamental passage from analysis to synthesis of chaotic properties, just based on the knowledge and exploitation of the dynamical systems theory deeply studied in the past.

Since then, control of nonlinear dynamics and chaos – intended in a broad sense – has become a hot trans-disciplinary research topic, as shown not only by the already mentioned great number of scientific publications but also by its increasing popularity within the environment of non-technical chaos connoisseurs.

In most of the works on the subject, the 1990 OGY paper is quoted along with some of the articles, aimed at improving and discussing the original idea or pursuing possible relevant applications, successively published by the same group of authors, with further collaborators, or by independent scientists. Without any attempt at completeness, one can mention, e.g., [42–47], in the first group, and [48–53] in the second group.

Yet, as already said, various underlying notions of chaos control stay in the background of the research activity in the field. In line with some first attempts, in their papers several authors aim at merely *removing* chaos, by means of classical control techniques [54], empirical methods [55], or other clever approaches [56]. These works are also referred to as “control of chaos”, although in this matter some authors, and we agree with them, prefer the name “suppression of chaos” (e.g., [36, 57–59]), which focuses on the effects of control rather than on the underlying skill of the control method. Thus, today, one refers to chaos control both when

1. chaotic transients and/or attractors are eliminated “tout court,” even if the tools employed to eliminate chaos have nothing to do with it, and when
2. typical properties of chaotic dynamics are involved in the control process, irrespective of the actual tools being employed to control the system and, indeed, of its actual response.

This twofold control aspect is already clearly highlighted in Linder and Ditto [36], when they assert that “... some techniques merely suppress or remove chaotic behavior ... others actually exploit chaotic behavior ...”.

Within the latter, more general and modern, perspective, the capability to exploit typical properties of dynamical systems undergoing chaotic behaviours in order to *control* the system response actually refers to either *suppressing* or *enhancing* chaos (the so called anti-control of chaos), or even *using* it, based on the

specific goal of the considered technique, on the system at hand, and on the area of interest [38].

Another basic issue is concerned with whether the dynamical phenomenon to be controlled by whatever “chaos control” technique has to be intended as “chaos” in a strict sense or, rather, as any kind of *complex behaviour* of a dynamical system, which may have different aspects according to whether a theoretical or a practical viewpoint is adopted. In the former, dynamical systems oriented, perspective, one can think of, e.g., any global bifurcational event possibly entailing complex behaviour of the system, of the relevant escape from a safe subset (a potential well or a basin of attraction) in parameter control space, or of the synchronization of different oscillators. In the latter, application-oriented, perspective, the kind and meaning of the “complex” phenomenon of interest is dictated by the practical goal to be attained.

In view of this extended notion of chaos control, no relevant exhaustive *classification* seems to be yet available, although some earlier interesting attempts have been made. For example, Chen and Dong [10] proposed a classification based on various tools employed in the control process (parameter-dependent approaches, open-loop strategies, engineering feedback control, adaptive control, intelligent control, etc.), while Fradkov’s “Chaos Control Bibliography (1997–2000)” [33] also contains a classification of the various applications in science and engineering. Recent classifications basically distinguish between feedback (or close-loop) and non-feedback (or open-loop) control techniques.

A more phenomenologically based classification relies on the ascertainment of how one can statically or dynamically modify either the system parameters – which represents the most basic form of passive control at the design stage – or the excitation, to attain the control goal. In particular,

1. the OGY’s method [9],
2. some classical methods (CM) of control theory [60],
3. the “control by system design” (CSD) [28],
4. the “parametric variation methods” (PVM) [10],

aim at somehow modifying the system. Note that also the OGY method could be classified as a PVM, but it has some specific features and we prefer to keep it separate.

In turn, methods acting on the excitation include

1. classical methods where a properly modified input (periodic or aperiodic, open-loop or feedback) is applied to the system (CM) [54, 56, 60, 61];
2. the “control through operating conditions” (COC) [28] based on modifying the frequency and/or the amplitude of the excitation;
3. the methods based on either combining parametric and external excitations (PEE) [41], or applying weak periodic perturbations (WPP) [17], or modifying the shape of the excitation (SE) [40, 62, 63].

Obviously, different methods are expected to give different performances, at least theoretically. Thus, one has methods aimed at:

1. stabilizing an unstable zone of parameter space (CSD, COC, PVM);
2. moving away from (previously known) chaotic zones (CSD, COC);
3. stabilizing a given, erratic solution (CM, OGY);
4. overall regularizing the system dynamics, irrespective of single solution behaviour (PEE, WPP, SE).

It is the authors' opinion that these latter distinctions are very important from a practical point of view because they suggest the use of the most appropriate control method fitting prescribed technical requirements.

Focusing the attention on the exploitation of chaotic behavior, there is even another possible classification, which seems not pursued in the literature, where various methods can be grouped according to the chaotic properties involved in the control. In this respect, one can distinguish methods based on

1. the properties of the saddles embedded in the chaotic attractor (OGY);
2. the ergodicity of the chaotic attractor (for example, all methods where a preliminary targeting step [46, 64, 65] is followed by the application of control tools in the neighborhood of the chosen control area belong to this class);
3. the sensitivity to initial conditions [45];
4. the occurrence of homo/heteroclinic bifurcations (PEE, WPP, SE).

Besides the formulation of different techniques for chaos control, the nearly two decades straddling the change of the Millennium have seen also a large number of relevant *applications* to mathematical, physical, biological, and engineering systems, within different technical fields. Of course, no attempt is made herein to comprehensively report on them. In contrast, consistent with one of the goals of this chapter, attention is focused on applications of chaos control in *mechanics* although, at a first sight, they could appear relatively few and of only minor importance.

Up to the authors' knowledge, earlier review papers explicitly devoted to this topic date back to 1993 [28]. The authors report on the newly developed (at that time) OGY method, and propose to control a mechanical system by either changing the forcing characteristics (e.g. the excitation frequency) – with the aim of moving away from a previously known chaotic region – or modifying any system property, such as inertia or stiffness. All of the proposed techniques fall in the area of “suppressing” chaos, and while being illustrated with reference to a Duffing oscillator (possibly equipped with also a tuned mass damper), they do not actually refer to specific issues associated with the mechanical nature of the systems.

A recent survey of the application of various methods of chaos control to mechanical systems [35] reports quite a long list of mechanical systems or processes for which control of nonlinear dynamics and chaos has been addressed in the literature, furnishing at least one reference for each of them. The list includes pendulums, beams and plates, systems with friction and/or impacts, spacecraft, vibroformers, microcantilevers, ship oscillations, tachometers, rate gyros, Duffing oscillators, robot-manipulator arms, earthquake civil engineering, milling processes, whirling motions under mechanical resonance, and systems with clearance. The items in the list highlight the quite scattered nature of the work in terms of the *mechanical*

complexity and variety of the systems, of the involved dynamical processes, and of specific control goals.

Indeed, choosing a proper framework within which to organize, present and discuss the state of art in the field is not an easy task because, depending on various application fields, mechanics is concerned (i) with both discrete (finite-dimensional) and continuous (infinite-dimensional) dynamical systems, (ii) with their possibly reliable low-dimensional models, and (iii) with a considerable richness of dynamical processes and phenomena to be possibly controlled. Though being mechanics mostly concerned with large scale systems traditionally analyzed and designed via linear techniques, a large amount of research made in the last 30 years has highlighted a cornucopia of nonlinear dynamic phenomena, including chaotic responses, along with their importance in mechanical and structural applications. Yet, regular nonlinear phenomena of such a variety of systems are already quite complicated in themselves, and as such exhibit a major importance in the real world of mechanics. As a matter of fact, the question of how important and pervasive chaotic phenomena are in the behavior of mechanical/structural systems, is still to be answered. Accordingly, it is quite difficult to select studies devoted to chaos control in mechanics in a strict sense, whereas the topic has to be intended in the wider sense of control of *dynamical complexity*, and is often addressed in the literature within the more general area of control of nonlinear, wanted or unwanted, phenomena.

On the other hand, the inherent complexity of systems from mechanical science and engineering poses the challenging problem of identifying proper reduced order models able to reliably describe the ensuing rich nonlinear dynamics; an issue which becomes even more demanding if one is interested in dealing with possibly complex nonregular response of the actual system and with its control. This basically entails one major consequence and one question.

1. A large majority of research works on control of chaos in mechanics is concerned with (relatively simple) archetypal nonlinear oscillators often representing idealizations of more involved discrete systems encountered in real mechanical engineering or minimal discretized representations – according to some reduction technique – of the infinite-dimensional continuous systems typical of structural engineering. This also entails that in most research papers the interest is focused on the control technique – or on other dynamical or control aspects transversal and unifying with respect to other scientific/technological fields – rather than on the actual mechanical peculiarities of the considered systems.
2. Even being convinced of the significance of nonregular dynamics – and of their control – for real mechanical/structural systems, how representative of their actual behavior are the chaotic regimes highlighted for such reference archetypal oscillators? Of course, this is a general point arising in any modeling and reduction problem, but it is felt to have a special meaning in mechanics owing to the above mentioned inherent complexity of the involved large-scale systems. The only possibility to clarify this issue consists in highlighting the actual occurrence, and the features, of chaotic response in experimental (mechanical and structural) systems, in developing refined theoretical and numerical models of the actual

system, and in cross-validating the two approaches. In this respect, considerable research is going on to highlight occurrence and features of chaos in experimental and/or refined theoretical models, but relatively little is yet done as regards their control.

Altogether, considering such a rich and intriguing framework which encompasses several aspects, the objective and organization of this chapter can be stated as follows.

The major role played by the OGY method in establishing the basic philosophy of chaos control is recognized in Sect. 2, along with its improvements, extensions and applications.

The use of OGY method for chaos control in mechanics is systematically addressed in Sect. 3, by identifying distinguishing classes of systems and applications, and by discussing them in the framework both of the capabilities and of some limitations of the method.

Then, a substantially alternative technique – according to the previously mentioned classification criteria – is presented in Sect. 4. It consists of an overall procedure for controlling the non-regular dynamics of systems triggered by the occurrence of homoclinic or heteroclinic bifurcations, which is discussed in its generic/non-generic features and with regard to a number of applications to systems of interest in nonlinear structural dynamics and engineering.

2 The Paradigmatic OGY Method for Chaos Control

The OGY method was first formulated in the celebrated work by Ott et al. [9]. A web-based search on “Scopus” looking back to the year 1998 (i.e., only 10 years back from now, instead of 18) already furnishes more than 2,300 articles citing the original OGY paper, with many of them also citing various other papers concerned with the method and published in the following years. In fact, the scientific literature in the area registered the publication of a great number of papers, co-authored either by one of the original authors with collaborators or by other scientists somehow related with the University of Maryland Chaos Group, for the whole decade of the Nineties. These papers provided explanations, clarifications, extensions and improvements on the theoretical and computational aspects of the method, as well as a great number of experimental implementations, applications in a wide variety of technical fields – ranging from mechanical systems to electronics, laser technology, chemical reactions, heart tissues, communications –, and modifications/refinements aimed at overcoming some observed limitations or drawbacks. The number of papers ensuing from the original one is so large that it makes no sense to list or report on them. A similar reasoning can be done as regards the theoretical and practical aspects of the method, which are somehow summarized in most – if not all – of the articles subsequently published by different authors and concerned with OGY control of chaos.

Quite soon the method became popular also within the less strictly technical literature, as witnessed by a number of articles appeared, e.g., in *Nature* [47], *Non-linear Science Today* [66] and *Physics Today* [67]. In both the hard and soft science respect, one could even run the risk to somehow compare, though to a lower scale, the paradigmatic role played by the OGY method in the area of control of chaos at the round of the Millennium with the revolutionary role played by chaos in the area of nonlinear science in the second half of the XX century!

Based on previous points, no analytical description of how the method works is reported herein. Yet, to the sole purpose of properly frameworking the discussion on the use of OGY control of chaos in mechanics, successively made in Sect. 3, a brief summary of the main qualitative features of the method is provided following the updated – though popular – exposition given in Ott [68].

Two fundamental aspects of chaos are:

1. The exponential sensitivity of orbits to small perturbations.
2. A very complex orbit structure making possible many different motions in the same system at fixed parameter values.

Actually, these are not independent properties, but rather two sides of the same “chaos coin”. The first property, typically quantified by the largest Lyapunov exponent, refers to the difficulty in the prediction of sufficiently distant future states of the system, while the second property, most often quantified by an entropy measure, refers to the fact that chaotic attractors often have embedded within them an infinite, dense set of unstable periodic orbits. While chaos is commonly viewed as undesirable, these properties can also be of great benefit in certain situations where one wishes to control the system dynamics. In particular, if the system is sufficiently well characterized and noise is not too large, then the first property implies that relatively large changes in the eventual location of orbit points can be made using only small changes in a control variable, while the second property implies that the dynamical changes so produced may be very diverse, leading to a situation with a great deal of flexibility. The two methods that are useful in utilizing chaos in an effective way are

1. *control*, namely the feedback stabilization of selected unstable orbits embedded in the attractor, and
2. *targeting*, namely the rapid steering of a chaotic orbit to the vicinity of a desired point in phase space.

Here *control* means *feedback control*, i.e., measurements of the state of the system are regularly taken, and, based on them, some controllable parameter (or set of parameters) is adjusted so as to achieve some goal. The desirable goals may vary, and different types of goals lead to qualitatively different control problems. Apart from those of interest for specific applications, general goals include:

1. Given a steadily running chaotic system, how can one improve its time averaged performance? Here performance is defined with reference to the specific function the system is meant to carry out.

2. Given a chaotic system in a given state at some specific time, how can one nudge the subsequent orbit to evolve rapidly from its current state to a different target location in state space ?

For the first goal, the complex orbit structure is most relevant; for the second goal, exponential sensitivity is most relevant. In all cases, the fundamental attributes of chaos imply that the control goals can be potentially achieved by use of only small controlling perturbations. Thus, one consequence of chaos is that control can be accomplished with low-energy/low-force controllers.

In order to see how this might be done, one first refers to the infinite number of unstable periodic orbits (UPOs) embedded within typical chaotic attractors. If one had the ability to perfectly place an initial condition on any chosen one of these UPOs, then an infinite number of different types of motion could be achieved. Moreover, if no noise were present in the system, such motions would ideally continue indefinitely in time.

The performance of a system, in general, depends on its state and its history. Let $x(t)$ denote the system state as a function of time t . Assume that the performance P of some steadily running process can be given as the time average of some quantity f that is a function of the system state $P = \langle f(x(t)) \rangle$, where the angle brackets denote a time average. For each UPO, a different state trajectory $x_i(t)$ results, where the subscript i labels the particular UPO. Consequently, each unstable periodic orbit i will have associated with it a performance $P_i = \langle f(x_i(t)) \rangle$, and these performance values will typically be different for each i . Moreover, it can be shown that, under suitable conditions, the performance P_c for the chaotic orbit is a weighted average of the performances P_i attained by the periodic orbits. The implication of this is that some of the P_i will be larger than P_c . If the system can be controlled to such a periodic orbit, then the system performance will be improved.

The next question is how to control the system so that it follows the chosen UPO rather than the chaotic orbit $x_c(t)$. One way of doing this is to wait until the ergodic uncontrolled chaotic orbit $x_c(t)$ comes close to the desired UPO, and then give it a small kick to place it on or very near the UPO. Due to several factors (system noise, our inability to kick the orbit to a precise location, an imperfect knowledge of precisely where the desired orbit lies), the kicked orbit will not be exactly on the desired UPO. Since the UPO is unstable, the system orbit will begin to move away from it. As soon as this is discerned, a small kick can be reapplied to reposition the system orbit closer to the desired UPO. By doing this continually, the orbit can be kept close to the desired UPO indefinitely. Note that for small noise and inaccuracy, the size of the kicks required to maintain the system orbit near the desired UPO is correspondingly small, approaching zero for the noiseless, absolutely accurate case. Thus, in many situations, the goal can be achieved with small controls. In comparison, one should note that in other proportional feedback control schemes where the targeted state might not be a part of the intrinsic dynamics, large controlling signals might be needed even if the system is noiseless.

Two issues that must be considered for such implementations are

1. how to determine and locate UPOs embedded in a chaotic attractor, and
2. how to make the small controlling kicks.

Determining UPOs If a very accurate analytical model of the system is available, then standard numerical techniques (e.g., Newton's method applied to the fixed point equation of the n times iterated map) can be applied to determine UPOs. However, in many cases of experimental interest, an analytical model of the system under study may not be available. In such cases, it is still possible to determine UPOs purely from data by recording the trajectory of a free-running (i.e., uncontrolled) chaotic orbit. The idea is to use state space embedding and attractors reconstruction techniques [69]. Techniques for finding UPOs from data have been discussed by So et al. [70] and by Pierson and Moss [71], among others.

Another issue is how many UPOs on the chaotic attractor need to be determined. Hunt and Ott [72] showed that it is seldom necessary to determine a very large number of UPOs since maximal performances typically occur on low period UPOs, which are limited in number.

UPO Controlling Algorithms Having chosen a suitable UPO embedded within the attractor, it remains to specify how the small controls should be programmed to maintain the system orbit on the chosen UPO. Several ideas have been proposed to accomplish this:

- The original OGY technique employed in Ott et al. [9] is to use the control to place the orbit on or near the stable manifold of the desired UPO.
- Romeiras et al. [73] discuss use of the "pole-placement" technique, which is standard in control theory.
- Dressler and Nitsche [74] and So and Ott [75] show how UPO control can be implemented using only time-series measurements of a single scalar state variable.
- Socolar et al. [76] present a technique particularly useful for the control of very fast dynamics.

Targeting refers to the control goal of quickly bringing an orbit to some desired location in state space. The basic idea is that since chaos is exponentially sensitive to small orbit perturbations, such orbit perturbations become large in a relatively short time. Moreover, if the perturbations are very carefully chosen, then there is the hope that the orbit can be efficiently directed to the target using only small controls. Several techniques for achieving this goal have been formulated [46, 77–79].

A number of improvements and modifications of the OGY control, aimed at overcoming some of the associated limitations, will be mentioned in the following section in connection with use of the method, or of any of its many variants, for control of chaos in mechanical systems. It will be shown how the method works well, both numerically and experimentally, for a considerable number of relatively simple systems and models. Yet, one major problem is concerned with the extent of the impact that the method can have for chaos control in real world applications.

This is linked with the general problem, mentioned in Sect. 1, of how much reliably representative of actual systems behaviour are the nonlinear dynamic phenomena and their control as evidenced for substantially archetypal models: in this sense, all possible limitations, if any, are not to be ascribed to the sole OGY method.

3 Use of OGY Method for Control of Chaos in Mechanics

Two features of a chaotic system make it a candidate for OGY control [80]: (i) the presence within the strange attractor of one or more unstable fixed points of the saddle-node type, to be possibly stabilized, and (ii) an accessible system parameter through which an applied control input changes the location of the unstable fixed point in the phase plane, adjusting it in such a way to ensure that the trajectory returns to the vicinity of the original one.

In theory, the OGY method is a “black box” technique that can be applied to any system possessing these two attributes, even to experimental systems for which no accurate analytical or numerical model exists. Indeed, it is not necessary to have a mathematical model to achieve the control goal since all control parameters may be resolved from time series analysis with the assistance of delay coordinate embedding techniques [81, 82]. Yet, the method does not seem to work well when a relatively large influence of noise exists as in usual experimental applications (see [83] for a feedback controlled pendulum), although it has been applied successfully to several experimental models [47].

Also, some features of the method, such as considering a local linear return map in the region around the unstable fixed point to be stabilized and applying small control forces to ensure that the oscillator continually returns to the region of phase space described by such a map, may render the control ineffective or impractical for some sensitive systems. This can occur, e.g., if the true dynamics change too quickly in the region of interest, if the region of linear validity is too small for some unstable fixed points, or if the unstable eigenvalue is so large that points near the fixed point are swept away very quickly and tend not to return to the linear region [84]. Moreover, since control only begins when the trajectory enters for the first time the region where the linear map applies, it can happen that for some chaotic systems this may take thousands of cycles, depending on the effect of the control parameter on the estimated location of the perturbed fixed point and on the size of the linear capture region. So, the OGY method, which is locally linear, may not be practical for all chaotic systems.

Another issue related to control of chaotic attractors is the existence of a sometimes long chaotic transient before the system settles onto the stabilized orbit. This transient arises because time-dependent parameters can only be effectively applied to stabilize an orbit when the system falls within a narrow region around it. For typical initial conditions the system wanders chaotically and since the attractor is ergodic it will eventually fall sufficiently close to the fixed point to be stabilized. The length of the chaotic transient for randomly chosen initial values has

an exponential distribution. To reduce the necessary time for stabilizing the target orbit, the OGY method was modified by Shinbrot et al. [46] (SOGY method) by exploiting the extreme sensitivity of a chaotic attractor to tiny perturbations in initial conditions to rapidly direct a system from a given initial state to a desired one. Overall, this targeting approach provides a way of directing trajectories from any initial conditions to the desired fixed point, and is shown to be quite successful.

There are other improvements of the OGY method aimed at overcoming some of its original limitations, as for example control of high periodic and high unstable UPO [85, 86] and control using time delay coordinates [74, 75, 87]. In this respect, it is worth mentioning how many of the presently available techniques aiming at controlling chaos by stabilizing UPOs represent some interpretation or extension of the original OGY method.

3.1 The Pendulum System

The mathematical pendulum is the archetypal nonlinear oscillator in mechanics. Accordingly, it has been considered for validating chaos control techniques, testing their effectiveness and comparing them. As regards OGY-based control methods, besides several numerical applications, also a number of experimental applications have been concerned with nonlinear pendulums [85, 87–89], as well as a double pendulum [90].

Starrett and Tagg [88] controlled chaos in a pendulum with vertically driven pivot over a range of driving frequencies and damping levels by using two variations on the OGY method, namely by proportional adjustments of damping for fixed time intervals or by proportional time intervals for fixed levels of damping, and with the eddy-current damping as control parameter.

Baker [91] studied the control of a chaotic damped driven pendulum (with one external excitation) by using OGY method for stabilizing an UPO through a feedback mechanism that periodically adjusts the damping parameter of the pendulum.

A number of papers/applications used some kind of “OGY-like” methods. For example, based on an optimal control algorithm “belonging to the parameter variation techniques introduced in [9]” and minimizing the distance between a chaotic trajectory and a desired periodic orbit [92], Bishop et al. [93] realized flexible control of tumbling chaos in a parametrically excited pendulum by stabilizing it onto a variety of oscillating or rotating periodic states, located via coupled symbolic-numerical dynamics, by small adjustments of the driving frequency.

Several authors compared the capability and efficiency of various techniques by referring to the (mathematical) pendulum, with the interest being focused rather on the former than on the latter.

Yagasaki and co-workers investigated chaos control for a pendulum subjected to feedforward and feedback control. Yagasaki and Uozumi [89] showed that the chaotic dynamics resulting from transverse intersection between the stable and unstable manifolds can be stabilized to the target saddle-type periodic orbit by

using OGY and SOGY methods. Since the former method required a very long time for stabilization of the targets and the latter was not effective when modeling errors existed, Yagasaki and Uozumi [48] developed another approach where nonlinear approximations are used for the chaotic dynamical system and for the stable manifold of the target. It also includes the OGY method as a special case and is no more than just the pole placement technique. Yagasaki and Uozumi [49] applied the method to the pendulum equation with the assistance of the delay coordinate embedding techniques [69], proving that the time necessary for stabilization is reduced, whereas Yagasaki and Yamashita [94] obtained a faster stabilization of the considered UPO without using delay coordinate embedding techniques, even when some modeling errors or the influence of noise exist.

Wang and Jing [95] applied the Lyapunov function method to design a controller able to convert the chaotic motion of a pendulum to any periodic orbit in a shorter time than that required by OGY-based methods (much shorter than OGY but also shorter than SOGY) for the same parameter values and initial conditions. In particular, the Lyapunov method is able to overcome a difficulty arising when employing even modified versions of OGY method, which make use of the generalized Poincaré map, for stabilizing higher-periodic orbits, namely the unclearly specified stability directions – which play a basic role in the control procedure – around a multiple saddle solution.

Alasty and Salarieh [96] designed a non-linear feedback controller to stabilize arbitrary desired periodic orbits such as period-one, period-two, period-four orbits and more in a chaotic pendulum, and showed that the major advantage of this method is its shorter chaotic transient time compared with other methods such as OGY.

A variation of the discrete OGY technique called semi-continuous control (SCC) method, proposed by Hübinger et al. [85] and extended by DeKorte et al. [87], was considered by Pereira-Pinto et al. [97] to stabilize unstable periodic orbits – identified by the closed-return method – in a nonlinear pendulum with torsional stiffness and damping. In implementing the control, two different situations were considered, first, if either all state variables are available, as it happens when considering signals generated by numerical integration of a mathematical model based on experimentally identified parameters, or, second, just a scalar time series obtained from an experimental setup. In the second situation, state space reconstruction is used with the method of delay coordinates [69]. Of course, experimental data is associated with noise contamination which is unavoidable in cases of data acquisition, so that the effect of noise on control techniques is an important point to be analyzed in order to ascertain their robustness, which is an essential aspect to the controllability of a dynamical system. By analyzing the effect of noise on the controlling procedure the authors defined some relevant limitations. Nonetheless, the possibility of using this approach to control chaotic behavior in mechanical systems was confirmed.

Working on the same line, Pereira-Pinto et al. [98] proposed the use of “extended state observers” – which is a tool from control theory for determining non-observed states of a dynamical system – to perform the state space reconstruction from one scalar time series, instead of using the delay coordinate method. Again, they suc-

cessfully applied this technique in the framework of the semi-continuous control of chaos of the nonlinear pendulum.

3.2 *Smooth Archetypal Oscillators*

Since the very first formulation of the OGY method, a number of authors considered some kind of magneto-elastic beam as the archetypal experimental model for control of chaos in smooth mechanics.

The practicality of the method to lock a system response onto specific periodic orbits embedded in a chaotic attractor and move it from one periodic orbit to another was first demonstrated by Ditto et al. [99] for a chaotically oscillating magneto-elastic ribbon, whose effective Young's modulus is sensitive to small magnetic fields. By providing a time-dependent perturbation to the applied field (based on the position of the beam), Ditto et al. could stabilize the ribbon onto period-1 and period-2 orbits.

Referring to the classical experimental magneto-mechanical oscillator consisting of a mass attached to a cantilevered elastic beam [100], which exhibits a double-well potential and a two-dimensional Poincaré map, Moon et al. [101] used the “occasionally proportional feedback” technique developed by Hunt [102] as an extension of the OGY method to stabilize the system onto a period-one motion.

Also stimulated by that classical experiment, Dressler et al. [103] demonstrated the tracking of an UPO in a horizontally cantilevered, periodically driven, elastic bronze ribbon. Stabilization of the UPO at each tracking step is performed via the local control method, which is a variant of OGY. Starting with feedback control vectors extracted from the analysis of the experimental data at each tracking step, the location of the UPO is redetermined using an adaptive orbit correction that exploits the applied control signal and the actual trajectory of the system. In this way, the UPO is tracked over a broad parameter regime where the chaotic attractor has disappeared and another periodic orbit has become stable.

A physical system made of a magnetoelastic metal ribbon was also considered by Ding et al. [104], along with a pair of coupled Duffing oscillators with periodic forcing, to demonstrate the viability in practical applications of a method of chaos control in high dimensions. Using small time-dependent perturbations of a single system parameter, the method achieves control by stabilizing a desired unstable periodic orbit with any number of unstable directions, which is a situation where the OGY method does not succeed. At the same time it is shown how, also in situations (e.g., just one stable and one unstable direction) where the low dimensional control is effective, a higher dimensional implementation may prove to be more efficient.

In turn, a number of numerically oriented works were devoted to controlling chaos in smooth nonlinear systems such as the escape and/or Duffing oscillators, via some variants of the OGY method.

Investigating the possibility to control chaos in a temporally irregular environment, Ding et al. [43, 105] considered the softening Duffing oscillator driven

by an irregular excitation as a model of ship rolling under lateral ocean waves. Heuristically incorporating a short-term monitoring and prediction feature of the future evolution of the environment into the OGY-SOGY control scheme, the authors were able to stabilize a period one orbit and prevent the ship capsizing that would occur, in absence of control, upon the boundary crisis of a chaotic attractor.

One interesting topic addressed in [54] is the comparison of the OGY method with respect to other control techniques, which is very useful because it permits one to judge the good/bad performances of the method, and the range of its applicability. More precisely, Sifakis and Elliott [54] considered four different control techniques applied to the classical Duffing oscillator: (i) open-loop periodic perturbation method, which consists in adding a periodic perturbation to the system excitation, (ii) continuous delayed feedback method, i.e., the Pyragas [56] method, (iii) the Hunt method [102, 106], and eventually (iv) the OGY method [9]. As to the first technique, in [54] the perturbation is chosen empirically by a trial-and-error procedure, whereas in other works it is optimally determined on the basis of a theoretical analysis relying on system dynamical properties [37].

3.3 Vibro-Impact and Friction Systems

Impact is often undesired in engineering systems, for it causes excessive fatigue, noise, or even direct failure. As a consequence, systems are designed to avoid operating conditions that are conducive to impact, where possible. Of course, the proper choice of system parameters at the design stage represents the most basic form of passive control. However, in many cases, the engineer is not given the opportunity to completely define and specify all system characteristics before manufacture. Unforeseen changes in the operating environment, damage from external forces, or modified system requirements can lead to unexpected and unwanted responses, such as a multiple impacting chaotic motion, to be possibly suppressed with a control technique. In fact, the reduction of impacting events is a potentially important practical benefit of using chaos control in the mechanical context [107].

Kalagnanam [108] applied the OGY algorithm to a spring-mass system that impacts with a sinusoidally vibrating table. Besides stabilizing its chaotic attractor on period-1 and period-2 orbits using small perturbations of the driving frequency, he demonstrated the ability to switch the chaotic system between the two orbits by controlled time dependent perturbations. Moreover, since the system exhibits long chaotic transients before settling onto the stabilized orbit, he exploited the exponential sensitivity of a chaotic system to small perturbations to control the duration of such a transient via the SOGY approach [46], which uses chaos to direct trajectories to targets and drastically reduces the average length of the chaotic transient.

Begley and Virgin [80] showed how, using the adjustable friction force as the control parameter, the linear map approximation of the OGY method may work in

a two-sided impact-friction oscillator for stabilizing either an ideally non-impacting response or a single-impact response corresponding to an UPO embedded in the chaotic attractor. Actually, while showing that, in mostly ideal cases, very little additional control effort (corresponding to low-energy effective control) is required for stabilization by the OGY method once the system locks onto the unstable orbit, they also discussed how in different practical situations the control of many other UPOs may be unattainable: this happens, for example, if a significant experimental noise is such as to push the system out of the linear region after control is established, or if large unstable eigenvalues, which are in fact common in impact systems due to substantial stretching of the phase space at impact, do occur.

One important point for vibro-impact systems is that the temporal evolution of the dynamical variables consists of smooth motions governed by a linear differential equation interrupted by a series of non-smooth impacts, with the trajectories being thus discontinuous in phase-space. Due to discontinuities, the hard part of the OGY control process, which applies only tiny perturbations on an available control parameter, is how to determine the value of parameter perturbation. For that, using the analytical solution of the differential equation and the impact rule, a transcendental (impact or discontinuity) map has to be determined, with the dynamical variables being computed at the impact instants. Referring to a fundamental discontinuity map with a square root singularity in the Jacobian, the so-called Nordmark map [109] that captures universal properties of impact oscillators near grazing, Casas and Grebogi [42] applied the OGY technique to select particular trajectories with a desired sequence of impacts and stabilized an unstable periodic orbit with one impact per period, involved in the grazing bifurcations, by adjusting slightly the parameter related to the external force. In turn, de Souza and Caldas [110], considering an impact oscillator or an impact-pair system with the amplitude of excitation as control parameter, used a transcendental map to specify the parameter perturbation needed to implement the OGY method for stabilizing an UPO embedded in the chaotic attractor.

Also for impact systems, a number of works were just explicitly “motivated” by OGY control or used some kind of “OGY-like” methods. Thus, Bishop et al. [111] developed a control strategy to stabilize period-1 impacting motions in an impacting driven beam undergoing complex motion associated with grazing bifurcations, as the driving frequency is varied. Galvanetto [112] adapted to non-smooth systems a controlling algorithm previously developed for smooth systems [92], and used it for finding UPOs of a stick-slip system made of a chain of blocks on a moving belt and for stabilizing them by small changes of one or more control parameters.

Motivated by the need to understand the nature of some nonlinear resonances in complex, multiactuated, servo-hydraulic structural engineering applications, Gutiérrez and Arrowsmith [113] analyzed the control of a mathematical model of the laboratory setup of a double-impacting system, studying the stability of low-order impact resonances in the context of the application of control techniques using displacement feedback. Preservation or annihilation of experimental and analytical resonant periodic orbits was realized via control schemes based on the OGY

method, showing how the dissipative nature of impact resonances makes it possible to apply a low-dimensional analysis.

It is also worth mentioning that vibro-impact systems with oscillating parts colliding with rigid walls or with other vibrating components may have important practical applications. This is the case of impact dampers, where the vibration of a primary system is controlled by the momentum transfer through collisions with a secondary loose mass which bounces back and forth [114]. Impact dampers are used to control high-amplitude oscillations, such as those typically occurring in chaotic motion, and hence can be regarded as devices for controlling chaos in mechanical engineering systems [115], as in cutting tools, turbine blades, and chimneys. No direct application of the OGY method seems to have been made in this respect, to the authors' knowledge. However, de Souza et al. [116] studied the control of chaotic impacts of a vibrating cart driven by a non-ideal motor, obtained by adding an impact damper consisting of a bouncing particle. The transfer of momentum which follows each collision between the particle and the walls attached to the cart is responsible for an effective coupling so that the motion of the cart can be controlled by the bouncing motion of the particle. With a very small value (about 0.5%) of the ratio between the masses of the bouncing particle and the cart, chaotic attractors are replaced by two stable trajectories with a kind of stabilization "resembling that achieved by the OGY technique". Of course, from the technical point of view, the smaller the ratio of the masses, the easier it is to install such impact dampers in engineering systems like turbine blades and cutting tools. The authors also pointed out how, with respect to previous studies [114], the value of this ratio is considerably lowered due to the non-ideal character of the forcing, which is a consequence of its limited energy supply.

Dry friction in mechanical oscillators can also lead to chaotic dynamics under periodic forcing. Moon et al. [117] demonstrated control and anti-control of chaos, namely stabilizing an UPO in a strange attractor or driving the system into a chaotic state near a periodic motion, in an experimental dry-friction oscillator obtained by adding lateral titanium friction plates to the cantilevered elastic beam with attached mass considered for control in Moon et al. [101]. Anti-control is often beneficial in engineering devices such as robotic manipulators or cutting tools for normal machining of metals, where it corresponds to adding small noise or dither onto periodic oscillations, e.g., to break static friction or to avoid hysteretic behavior. Control and anti-control were based again on the "occasionally proportional feedback" technique [102], and were effected by changing the normal force of the dry friction element via a magnetic actuator.

3.4 Coupled Mechanical Systems

A number of coupled systems have been studied for control by using OGY-like methods.

Considering a kicked double rotor system made of two connected massless rods, Feudel et al. [118] exploited its ability to access many different coexisting states,

combined with its sensitivity and flexibility, to gear the dynamics toward a specific periodic behaviour. They used a simple feedback control scheme making use of perturbations smaller than those of the standard OGY, and also smaller than the noise amplitude.

In a tutorial paper devoted to controlling chaos in a dynamical system constituted by two coupled dynamos, Agiza [119] also dealt with suppression of chaotic behavior to one of the UPOs embedded within the attractor but this was realized by nonfeedback or a delay feedback control method which does not require finding the UPO to be stabilized but needs its period only.

Control and stabilization of an UPO of a chaotic system using a fuzzy system constructed on OGY method were investigated in Alasty and Salarieh [96], who applied the method to a Bonhoeffer-van der Pol oscillator representing a model of a two degree-of-freedom system kept in a fluid flow. Attention is focused to the advantages to be possibly obtained via fuzzification rather than on the application. It turns out that the transient response time and the control effort for stabilizing the period-4 unstable orbit with fuzzy-OGY controller are much less than with pure OGY control. This is due to the fact that when using fuzzy control the perturbation signals applied to the system are not sensible to the distance from the fixed point of the periodic trajectory in phase space. Fuzzification of OGY perturbing signal causes a smoother perturbing signal in larger domains of time and Poincaré section. The main role of a fuzzy system is to make a control perturbing signal in the points which are far from the fixed point, while in the near vicinity of fixed point the perturbation made by the fuzzy system is similar to that made by the pure OGY controller. Robustness of controlled system against random disturbances also increases with fuzzy-OGY controller. Comparable favorable features are exhibited, to a certain extent, by a fuzzy-Pyragas controller with respect to the pure Pyragas one.

A number of mechanical systems are actually high-dimensional, as it happens, e.g., in robotics. Underactuated manipulators arise in a number of important applications such as free-flying space robots, hyper-redundant manipulators, snake-like robots, and manipulators with structural flexibility, among others. An important goal in manipulator control is to perform tasks involving the exact tracking of some desired trajectory. After a proper analysis of the particular device dynamics, exact tracking depends on the nature of the designed control algorithm. The absence of an actuator transforms the robot into an underactuated device that may lead to a malfunction of the system, to an erroneous tracking of the desired trajectory and in some cases to instability. Moreover, underactuated robots may exhibit rich dynamical behaviors including chaos [120, 121].

In addressing these issues, González-Hernández et al. [120, 121] dwelled on some limitations of earlier control techniques for stabilization of periodic orbits, including the OGY one, which are designed for systems described by three differential equations yielding two-dimensional Poincaré or Lorenz maps and are only capable of imposing a particular dynamics for stabilization. Overall, they noticed how a general framework for control of higher-dimensional systems is not yet established.

In their paper, the authors aim at controlling higher-dimensional systems and at imposing a desired dynamics for stabilization. They report on the use of the Lorenz, instead of the Poincaré, map for the local identification of the dynamics around the UPOs in a chaotic attractor, claiming that its use allows for better identification schemes and avoids ill-conditioned identification problems [122]. As a first stage for achieving stabilization of periodic orbits embedded in higher-dimension chaotic attractors, based on system measurements, they propose a framework for designing flexible control laws which still belongs to the class of parameter perturbation methods. Flexibility means the capability to impose arbitrary dynamics with the proposed control law, which is applied to the so called Pendubot, namely an underactuated robotic system with two links. The proposed method offers a way to control chaotic systems without any prior knowledge of their dynamics, and needs a measure of only one of the system variables and the availability of a system parameter which can play the role of a control input.

3.5 Targeting in Astrodynamics

Besides strict suppression of chaos, one important outcome of the OGY approach to chaos control has been the flexible and efficient exploitation of the complexity inherent in the chaotic dynamics for their technological uses. In particular, the inherent exponential sensitivity of chaotic time evolutions to tiny perturbations has been exploited for targeting, i.e. for rapidly directing a system to a desired accessible state [46]. Major relevant achievements have been obtained in the field of astrodynamics, particularly related to the important and timely issue of spacecraft transfer, to be realized with a low consumption of propellant or energy.

In general, controlling and targeting in chaotic Hamiltonian systems – which are the situations faced in astrodynamical problems related to spacecraft transfer – are not easy to accomplish. Besides the coexistence of interwoven chaotic and quasi-periodic regions, the phase space is divided into layered components, which are separated from each other by Cantori [123]. Typically, a trajectory initialized in one layer of the chaotic region wanders in that layer for a long period of time before it crosses the Cantori to wander in the next layer. Several targeting strategies have been proposed to overcome these difficulties, e.g., using small perturbations [124] to drive a trajectory from an unstable periodic orbit located inside a chaotic scattering region to a target point outside it, which is a major problem [125], and then applying in the “controllable region” a method of control of chaos derived from OGY [126]. Macau [127] applied the overall procedure to a Hamiltonian system in a “soft” chaotic Hamiltonian evolution, i.e., targeting in the planar, circular, restricted three-body problem which models the dynamics of a spacecraft moving in the Earth-Moon system and is a special case of the full three-body problem, where one of the masses is taken to be infinitesimal and has no influence on the two primaries which are on circular orbits.

A review of the relevancy and efficiency of the OGY strategy of chaos control to spacecraft steering in different Hamiltonian situations is presented in Macau and

Grebogi [44], who also reported on the Hill problem governing the encounter phenomenon of two light bodies describing circular orbits around a heavy central body. In this case, the unstable periodic orbits embedded in a non-attracting chaotic invariant set related to chaotic scattering are exploited for in-orbit stabilization, namely to keep one satellite in orbit around the other [128].

It is worth stressing the importance and efficiency of control of chaos and targeting techniques as regards low-energy spacecraft transfer and in-orbit stabilization. Indeed, since the lifetime of an automatic space exploration mission is mainly defined by the amount of fuel it can carry, the direct benefits of chaotic transfer in astrodynamics are twofold, i.e., either allowing a spacecraft to go farther in exploring the limits of our Solar System, or saving the amount of fuel, which entails a larger amount of scientific instrumentation in the payload, since the liftoff capacity of a launch vehicle is limited. These techniques can also be used to rescue spacecrafts that end up in a wrong trajectory accidentally, due to defects in the launch phase.

3.6 Atomic Force Microscopy

Atomic force microscopy (AFM) is widely used for nano-scale material characterization and surface inspection in engineering applications. Indeed, as an imaging tool, AFM is capable of resolving surface features at the atomic level for conducting and non-conducting samples, and it is currently used in many imaging applications ranging from biological systems to semiconductor manufacturing.

The mechanism of AFM basically depends on the interaction of a micro-cantilever with surface forces. The tip of the micro-cantilever interacts with the surface through a surface-tip interaction potential. One approach to measure the surface forces is to monitor the deflection of the micro-cantilever through a photodiode. This approach is named “contact mode”. Another approach termed “tapping mode” is performed by vibrating the micro-cantilever close to its resonance frequency, at a constant driving amplitude, and monitoring the changes in its effective spring constant.

As reported in the introduction of Arjmand et al. [129], a micro-cantilever in tapping mode may exhibit chaotic behaviour in some regions of its physical parameters (damping value, excitation amplitude and frequency, average tip-sample distance), as observed experimentally [130] and verified theoretically [131] and numerically [132]. Yet, chaotic behavior is highly undesirable for AFM performance because it entails inaccurate measurements and low resolution of the achieved sample topography. Accordingly, it is always required to eliminate the possibility of chaotic motion of the micro-cantilever either by changing the AFM operating conditions to a region of parameter space where regular motion is assured or by designing an active controller that stabilizes the system on one of its UPOs. Arjmand et al. [129] developed a nonlinear delayed feedback control algorithm to stabilize the AFM micro-cantilever on its first-order UPO, as well as a delayed feedback control via sliding mode scheme for chaos elimination.

In turn, Misra et al. [133] documented a variety of dynamical systems-based, control-theoretic analytical, computational and experimental tools for exploiting the natural oscillating dynamics of the cantilever in tapping-mode operation of AFMs. In particular, they applied different OGY-based feedback strategies to the AFM cantilever dynamics described by a lumped-parameter model in which the spatial shape of the vibrating beam is approximated by its first fundamental mode of vibration. By imposing discrete changes in the vertical offset between the cantilever support and the sample surface based on an estimated linearization of the system dynamics about a dynamically generated reference trajectory, they aimed at maintaining a desired oscillation amplitude of the cantilever and, indirectly, at extracting information during surface scanning of the sample profile. Three different control schemes, full-state feedback, estimated-state feedback and partial-state feedback, are examined in terms of their capability of stabilizing unstable periodic oscillations, reducing the extent of transient dynamics in the vicinity of stable periodic oscillations, and maintaining a desired periodic oscillation while scanning the sample surface at fairly high lateral speeds, under the influence of a train of disturbance inputs in the form of surface profile variations. Numerical results document increases in the speed of surface scanning, while preventing undesired transitions between different cantilever oscillations, as well as successful use of (in the absence of control) unstable cantilever oscillations that achieve tapping-mode operation with relatively small contact velocities and short duration of contact.

4 An Alternative Unified Framework for Control of Non-regular Dynamics of Mechanical Systems

One characteristic of the OGY method is that it *exploits* the chaotic properties of systems to achieve some desired performances, which in the specific case is the stabilization of an unstable saddle embedded in the chaotic attractor.

There are several other methods which likewise exploit the chaotic performances of dynamical systems to obtain certain goals, although they refer to different features with respect to the OGY method. Among them, an important role is played by the methods aimed at eliminating, or shifting in parameters space, a given homoclinic or heteroclinic intersection which triggers unwanted dynamical phenomena [17, 37, 41, 134].

The objective of eliminating the homoclinic bifurcation can be reached in various manners, namely,

1. by adding a controlling parametric excitation to an uncontrollable external excitation, or viceversa [41];
2. by modifying the excitation by adding controlling terms to a given uncontrollable one. This can be done, for example, by adding single or multiple subharmonics or superharmonics (or even ultra-subharmonics) to a given harmonic term [17, 37], and even by choosing these added terms in an optimal way [37];
3. by modifying the parameters of the system.

The last case is trivial from a theoretical point of view, and it is mentioned only because it can have some interest from a practical point of view. It just requires one to identify the chaotic region in the parameters space, and to change the parameters to stay outside that region.

The first two cases, on the other hand, are not very different from a conceptual point of view. The differences are mainly (but not uniquely) technical: the common idea is that of adding controlling terms to eliminate a given, unwanted, homo/heteroclinic intersection. So, we can describe just one of them, and we will illustrate the technique developed by Lenci and Rega in a series of papers [26, 37, 62, 63, 107, 135–146].

Any kind of control method based on the elimination of homoclinic or heteroclinic intersections, or better on shifting or eliminating the global bifurcations through which complex dynamics appear, requires various steps, both of theoretical and practical nature. Although they can be supported by analytical arguments, herein they are mostly discussed at a phenomenological level with the aim of introducing the general ideas and features of the method, usually not reported in regular publications, and of highlighting its overall framework. Of course, the various technical aspects can be found in the specific papers.

4.1 Single Degree-of-Freedom Systems

With the aim of capturing the key features without redundancies we refer to single degree-of-freedom (d.o.f.) models, which describe the dynamics of simple mechanical systems or of reduced order models of complex structures.

This choice may appear very restrictive at a first glance, but it is not. In fact, in general, the additional structure due to the higher dimensionality does not add any significant element to the control *ideas* and to the theoretical apparatus, but contributes mainly *technical* difficulties, which we want to by-pass because they are not due to the control, or can possibly add new resources, thus enlarging instead of reducing the applicability of control.

For example, in large dimensions, it may be difficult to practically measure the distance between stable and unstable manifolds, which is necessary to detect the considered bifurcation. This implies that it may be difficult to *practically* apply control to those structures, but this does not affect the *theoretical* characteristics of control, and the possibility of its application. What can instead occur is to take advantage of the complexity of the structure to look for further properties of the control, which add to those investigated with single d.o.f. models.

According to the previous considerations, the generic system which will be considered is governed by the ordinary differential equation

$$\ddot{x} + f_1(x) + f_2(x, \dot{x}) + f_3(x, t) = 0, \quad (1)$$

whose solution $x(t)$ represents the status of the system. Four terms, which are the main “components” of a common mechanical system, occur in Eq. (1): inertia, $\ddot{x}$,

conservative elastic restoring force, $f_1(x)$, dissipation or damping, $f_2(x, \dot{x})$, and external or parametric excitation, $f_3(x, t)$. Note that in general the term $f_2(x, \dot{x})$ may also provide a negative damping, i.e. it can pump energy into the system. However, in this work, we will consider only standard mechanical systems where $f_2(x, \dot{x})$ dissipates energy.

When f_2 and f_3 vanish, the system (1) is Hamiltonian, or “conservative”. In this case, the Hamilton function $H(x, \dot{x}) = \dot{x}^2/2 + \int f_1(x)dx$ is constant along orbits. In mechanical language H is the total energy, sum of the kinetic $K(\dot{x}) = \dot{x}^2/2$ and potential $V(x) = \int f_1(x)dx$ energy. The terms f_2 and f_3 represent two different kinds of modifications, or perturbations, of an Hamiltonian system: those which extract energy from the system, and those which pump energy into the system, respectively.

More difficult situations can of course be found in the reality, but the case (1) is sufficient for our purposes.

4.2 Different Kinds of Global Bifurcations

The first consideration, less obvious than what might be thought, concerns the relevance of the global bifurcations for the complex behavior. In fact, we can have two different situations: bifurcations not directly related to a specific outcome in the system dynamics, and bifurcations immediately causing a strong variation in the response.

While actually being the same dynamical phenomenon, there are strong differences between the two classes of bifurcations. The former usually triggers chaotic mechanisms, such as, for example, the creation of a chaotic saddle, which do not immediately, and dramatically, influence the response, but which constitute the necessary pre-requisites for successive routes to chaos. Their effects are hidden behind the visible outcome, but they are not negligible, even from a practical point of view.

Remarkable examples of this case are the homoclinic or heteroclinic orbits of hilltop saddles of Hamiltonian systems. Their invariant manifolds usually surround potential wells where the important part of the dynamics occurs, and constitute a barrier against the penetration of trajectories from out-of-well attractors eroding the well. Once this barrier has been broken by the global bifurcation, the (chaotic) effects can penetrate the well and, sooner or later, become visible. From the one side, this permits deeper understanding of the dynamical mechanisms involved in the event, especially with respect to safe well erosion [146, 147], and from the other side it permits detailed and specific investigations, to be followed by smart application of the control method.

Due to the underlying Hamiltonian nature, these bifurcations can be usually detected analytically by the Melnikov method [48] or even directly in closed form. This is important because it permits a more general application of the control method [62, 137].

The second class of bifurcations is involved in specific boundary or interior crises, and determines sudden appearance/disappearance or enlargement of chaotic

attractors or sudden modifications of their basins of attraction. Thus, their elimination directly entails elimination of the crises, with immediate, visible, effects [136]. These are the situations where the class of control methods discussed in this section are very effective. Here in fact even a small control leads to remarkable results, and, e.g., the chaotic attractor can be easily removed/created.

The drawback consists in the fact that these bifurcations are rarely related to hilltop saddles, and commonly cannot be easily detected or even discovered. In fact, it can (and actually does) occur that the saddle S directly responsible of a crisis is hidden by other, possibly more robust, saddles with an heteroclinic connection with S . This can generate an heteroclinic tangle [148], which makes any attempt to detect the saddle and its invariant manifolds a challenge from a practical (i.e. numerical) point of view, and even a somehow questionable theoretical issue.

In the most fortunate cases where, in spite of previous difficulties, the detection is possible, it usually requires long, time consuming, and systematic numerical investigations, which are cumbersome and do not permit a full understanding of control or its full exploiting. Thus, the application of control is less general than in the former case.

However, it should be clear that, strictly speaking, the detection of the saddles and of their manifolds, which is a necessary pre-requisite for applying control, is not a part of the control method, which can actually be developed, in its general lines, without any reference to a specific situation (and even, in some cases, without references to any specific systems). The ability of detecting the global bifurcations directly responsible for the observed crises simply provides a more powerful possibility of eliminating/creating chaos by a straightforward application of a control method.

The previous considerations explain why we will deal with homoclinic or heteroclinic bifurcations of hilltop saddles (but see also [136] where the control is applied to a numerically detected homoclinic bifurcation of a non-hilltop saddle). In fact, although they are involved in boundary or interior crises only in some situations, they are relatively easy to be analytically detected by the Melnikov or similar method, thus permitting a detailed development of the most important aspects of control, without reference to any specific situation. To support this choice, we further underline that the considered global bifurcations trigger the dynamical events eventually leading to chaos, so that its elimination is useful in any case.

4.3 Distance Between Stable and Unstable Manifolds

From a practical point of view, the key to applying any kind of control based on the elimination of homoclinic or heteroclinic bifurcations is the detection of distance between the stable and unstable manifolds.

To illustrate the main properties, we refer to an homoclinic bifurcation of a nearly Hamiltonian system (Fig. 1), i.e., a conservative system with small perturbations, which is supposed to have a saddle point with an homoclinic orbit. The major aim is that of verifying whether the homoclinic orbit survives or not when damping and

excitation, here seen as perturbations, are added. The objective of this section is indeed determining the distance between perturbed stable and unstable manifolds.

In the absence of perturbations, the stable and unstable manifolds of the saddle coincide because they are constrained to have the same energy (Fig. 1a).

4.3.1 Effect of Damping

When damping is added to the Hamiltonian system, the coinciding stable and unstable manifolds – which still exist – split away from each other (Fig. 1b). In fact, the saddle does not change energy (being a fixed point), the unstable manifold loses energy in forward time (and asymptotically approaches an attractor different from the saddle), and the stable manifold increases its energy backward in time (as it must approach the saddle as $t \rightarrow +\infty$ and its energy decreases forward in time). Thus, a homoclinic loop is no longer energetically possible and, on appropriate Poincaré maps, usually the stroboscopic one, there is a non zero distance between stable and unstable manifolds. It is easy to realize that this distance is proportional to the damping amplitude, herein denoted by $\varepsilon\delta$, and that in the case of small damping, it is approximately constant along a sufficiently short interval (Fig. 1b).

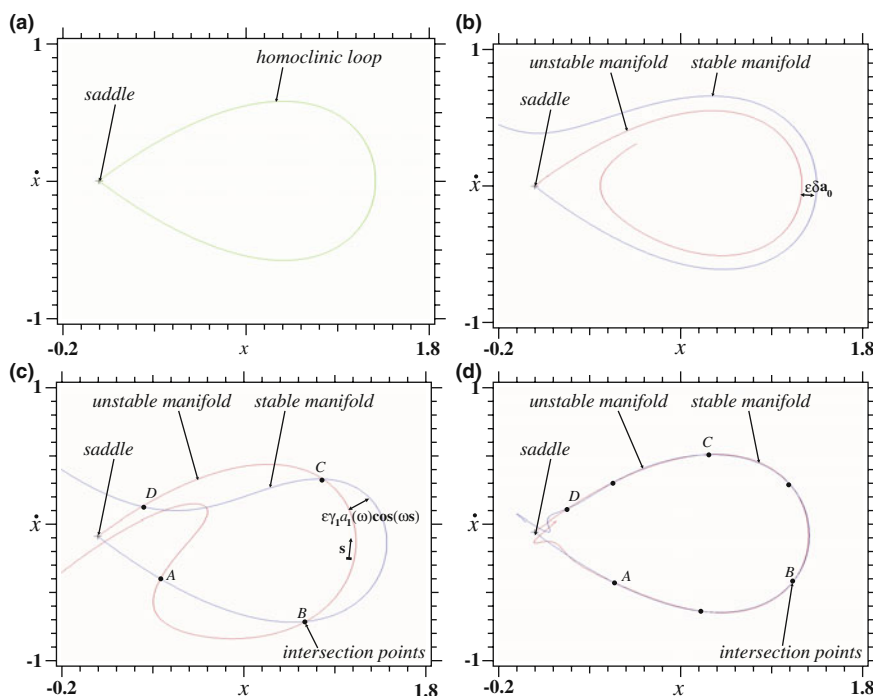


Fig. 1 Schematic illustration of the distances between stable and unstable manifolds in an homoclinic bifurcation of a nearly Hamiltonian system. (a) Hamiltonian case; (b) damped case; (c) harmonically excited case, excitation frequency = ω ; (d) superharmonically excited case, excitation frequency = 2ω

4.3.2 Effect of Excitation

The second standard perturbation is the excitation $f_3(x, t)$, which is assumed to be periodic in time with angular frequency ω and zero mean value (a constant excitation simply shifts the reference/equilibrium point of the system). Furthermore, it is initially supposed to be harmonic and state independent or “external,” namely, $f_3(x, t) = \varepsilon \gamma_1 \sin(\omega t)$, $\varepsilon \gamma_1$ measuring the excitation amplitude. The extension to state dependent “parametric” excitations can be obtained by referring to the so called *equivalent external* excitation in the evaluation of the manifolds distance, as shown in [137].

Both the definitions of the amplitudes $\varepsilon \delta$ and $\varepsilon \gamma_1$ are left somehow vague up to now, but the physical intuition is sufficient for the purposes of the present section. We only stress that they are small by the introduction of the dimensionless smallness parameter ε .

The excitation affects the invariant manifolds by producing an oscillating distance on the Poincaré map, as shown in Fig. 1c, where some of the primary intersection points (P.I.P., see [148]) are marked by A , B , C and D . Note that, according to common sense, the wavy pattern of the excitation produces a wavy pattern of the distance, and this is confirmed by Fig. 1d, which shows that doubling the frequency of the excitation doubles the number of P.I.P.

For what concerns the magnitude of the distance, it is clear that it is directly proportional to the excitation amplitude, and that for small amplitudes it depends linearly on it. Moreover, comparison of Figs. 1c and 1d shows that the magnitude of the distance also depends on ω , and in the particular case of Fig. 1 it (rapidly) decreases with increasing ω . In other cases it can instead increase with ω , but anyway *the magnitude of the distance strongly depends on the excitation frequency*.

4.3.3 The Perturbed Manifold Distance

Based on the previous considerations, and focusing attention on the interval $A - B - C - D$ of Fig. 1c, d, we can write the distance between stable and unstable manifolds in the following approximate way (see Fig. 1b, c):

$$d(\omega s) = \varepsilon \delta a_0 + \varepsilon \gamma_1 a_1(\omega) g(\omega s), \quad (2)$$

where s is an arc length parameter (Fig. 1c), a_0 and $a_1(\omega)$ are a constant and a function depending on the specific dynamical system, and ε stresses that the assumed linearity with respect to δ and γ_1 holds only for small damping and excitation amplitudes. The smallness is also tacitly assumed in the additive form of the distance. Finally, in Eq. (2) $g(\omega s)$ is the oscillating part of the distance. It is due to the periodic excitation, and like it, it has zero mean value and is periodic with the same frequency. Since the excitation is harmonic and with small amplitude, it is possible to approximate $g(\omega s)$ with an harmonic function, e.g., $g(\omega s) = \cos(\omega s)$. Using cosine instead of sine has an unessential technical reason, linked to the choice of the point for which $s = 0$ (Fig. 1).

Equation (2) captures two opposite effects suggested by the physical intuition, according to which damping gives a positive distance, thus contributing to manifolds non-intersection, whereas harmonic excitation gives an oscillating distance, thus contributing to manifolds intersection.

4.3.4 More General Excitation and Damping.

The extension of the previous consideration to the case of a non-harmonic, but still periodic and external, excitation

$$\begin{aligned} f_3(x,t) &= \varepsilon \sum_{j=1}^N \gamma_j \sin(j\omega t + \Psi_j) = \\ &= \varepsilon \gamma_1 \sum_{j=1}^N \left(\frac{\gamma_j}{\gamma_1} \right) \sin(j\omega t + \Psi_j) = \varepsilon \gamma_1 f(\omega t), \end{aligned} \quad (3)$$

is straightforward. In fact, by using the linearity, consequence of the smallness of the amplitudes, we have that

$$\begin{aligned} d(\omega s) &= \varepsilon \delta a_0 + \varepsilon \sum_{j=1}^N \gamma_j a_1(j\omega) \cos(j\omega s + \Psi_j) = \\ &= \varepsilon \delta a_0 + \varepsilon \gamma_1 a_1(\omega) \sum_{j=1}^N \frac{\gamma_j a_1(j\omega)}{\gamma_1 a_1(\omega)} \cos(j\omega s + \Psi_j) = \\ &= \varepsilon \delta a_0 + \varepsilon \gamma_1 a_1(\omega) h(\omega s), \end{aligned} \quad (4)$$

More generally, the extension to arbitrary – but small – damping $f_2(x, \dot{x})$ and excitation $f_3(x, t)$ leads to an expression of the distance with the same structure as (4). The differences are only of technical nature, and appear in the expressions of a_0 and $a_1(\omega)$, and therefore in a different relation between the Fourier coefficients of the excitation and those of the distance between the manifolds. In fact, it is not difficult to accept that a generic periodic excitation gives an oscillating distance, and that the shape of this oscillation is due to the shape of the excitation in a way which can be computed, at least in principle. Therefore, from now on we will refer directly to (4) without specifying the nature of the system, of the damping and of the excitation, which indeed are “hidden” behind (4). A schematic representation of this distance is reported in Fig. 2. The constant part due to damping and the oscillating part due to the excitation are clearly shown.

Equations (3) and (4) are written in the last forms because we assign $\varepsilon \gamma_1$ the role of overall excitation amplitude, while the remaining functions $f(m)$ and $h(m)$ measure the shape of the excitation and its effects on the manifolds distance. Note

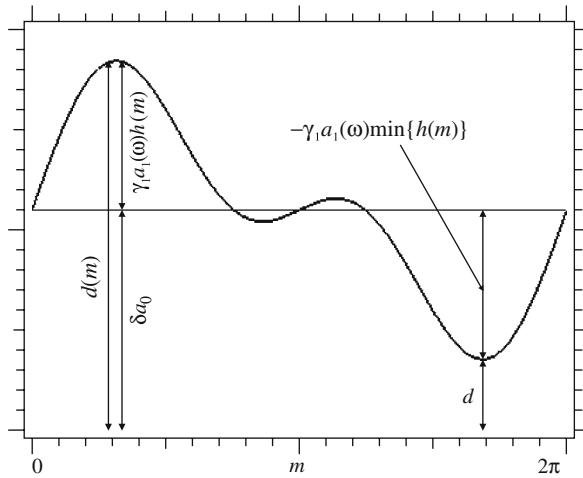


Fig. 2 The schematic illustration of the distance between stable and unstable manifolds

that, by definition, both $f(m)$ and $h(m)$ are 2π -periodic and $f_1 = h_1 = 1$, while $h_j = \frac{\gamma_j a_1(j\omega)}{\gamma_1 a_1(\omega)}$ (f_j and h_j are the Fourier coefficients of the respective functions). Furthermore, Ψ_1 represents an unessential phase shift, and can be assumed equal to zero without loss of generality (it is sufficient to consider $\hat{\Psi}_j = \Psi_j - j\Psi_1$ instead of Ψ_j).

4.3.5 Reference (Natural) and Controlling Excitations

The previous assumption of considering a general (although still periodic) excitation is important for the development of control, and is now briefly discussed. In fact, if one wants to operate by modifying the excitation, for example, by adding to a given natural excitation a properly designed controlling excitation, as done in classical control [60], one must have clearly in mind that the natural excitation cannot be touched, otherwise the problem becomes trivial. In other words, one must have a reference, unchangeable, excitation with respect to which to measure the improvements due to the control. On the contrary, it is trivial to change the excitation in order to get a desirable dynamical response: it is sufficient to look at the behavior chart of the system and choose the excitation (or its parameters) which provides the desired response. E.g., whatever the reference excitation is, if we are able to change its amplitude it is obvious that we can easily move from chaotic to non-chaotic response regimes.

In this chapter, where only periodic excitations are considered, the reference is the *harmonic* excitation, which is the most natural and intuitive. Thus, we implicitly think of the excitation as made of two parts,

$$f_3(x,t) = \varepsilon \gamma_1 \sin(\omega t) + \varepsilon \hat{f}_3(x,t) = \varepsilon \gamma_1 \{\sin(\omega t) + \bar{f}_3(x,t)\}, \quad (5)$$

In general, the two parts have their own amplitudes, but we find it useful to refer to a unique excitation amplitude, as done in the last expressions of (3), (4) and now in (5), where the overall excitation is just that of the reference excitation. Other choices are possible in principle, but the present one seems to be the most natural in the field of the proposed control method.

Within the framework of periodic excitations, the choice of the harmonic one as the reference excitation is not restrictive. In fact, if another reference excitation is chosen, then Eq. (5) becomes

$$f_3(x,t) = \varepsilon \gamma_1 \{f_{3,\text{ref}}(x,t) + \bar{f}_{3,\text{cont}}(x,t)\}, \quad (6)$$

where once more the amplitude of the reference excitation has been singled out and interpreted as the overall amplitude. By equating the two expressions of $f_3(x,t)$ given in (5) and (6) we get,

$$\bar{f}_{3,\text{cont}}(x,t) = \sin(\omega t) - f_{3,\text{ref}}(x,t) + \bar{f}_3(x,t), \quad (7)$$

so that if we know the control excitation $\bar{f}_3(x,t)$ in the case of harmonic reference excitation, then we know the control excitation $\bar{f}_{3,\text{cont}}(x,t)$ corresponding to any reference excitation $f_{3,\text{ref}}(x,t)$.

4.3.6 Energetic Derivation of Perturbed Manifolds Distance

The expression (4) is previously obtained by qualitative considerations. In spite of this, it provides exact information, because it gives the same manifolds distance as computed by the Melnikov method [148]. Furthermore, it can be derived by energy arguments as follows (see [149] for similar reasonings).

An (hypothetic) homoclinic orbit $x_h(t)$ is possible if and only if the sum of the total energy E_d dissipated by damping and of the total energy E_p pumped by the excitation is zero. These energies are given by, respectively,

$$\begin{aligned} E_d &= \int_{-\infty}^{+\infty} f_2[x_h(t), \dot{x}_h(t)] \dot{x}_h(t) dt, \\ E_p &= \int_{-\infty}^{+\infty} f_3[x_h(t), t] \dot{x}_h(t) dt. \end{aligned} \quad (8)$$

Let us define the “energetic distance” $d = E_d + E_p$, so that a condition for $x_h(t)$ to be an homoclinic orbit is

$$d = \int_{-\infty}^{+\infty} f_2[x_h(t), \dot{x}_h(t)] \dot{x}_h(t) dt + \int_{-\infty}^{+\infty} f_3[x_h(t), t] \dot{x}_h(t) dt = 0. \quad (9)$$

Remark. An alternative, but equivalent, way to obtain (9) is the following. Multiply the field equation (1) by $\dot{x}_h(t)$ and integrate in time from $t = -\infty$ to $t = +\infty$. We get $H(+\infty) - H(-\infty) + d = 0$, where H is the Hamilton function. Since the orbit is

homoclinic, it approaches the same saddle for $t = \pm\infty$, so that $H(+\infty) = H(-\infty)$ and $d = 0$ follows.

If $d \neq 0$ the guessed $x_h(t)$ cannot be an homoclinic orbit. Recalling [148] that an homoclinic intersection between stable and unstable manifolds implies the existence of an homoclinic orbit, it is possible to conclude that if we are able to show that no solutions of (1) exist satisfying (9) and approaching the same saddle at infinity, then the manifolds keep disjoint and there is no homoclinic intersection. In other words, there is a link between the manifolds distance and the energetic distance, as they simultaneously vanish in the case of homoclinic intersection. By varying a system parameter, an homoclinic bifurcation occurs when both become zero for the first time.

The Eq. (9) holds in general, and it is an useful check for a guessed homoclinic solution. We specialize it to the considered case of a nearly Hamiltonian system. The unperturbed (Hamiltonian) system has an homoclinic orbit $x_0(t-s)$ (see Fig. 1), where t is the real time and s an arbitrary phase shift introduced because the reference system is autonomous (see Eq. (1) with $f_2 = f_3 = 0$). Let the perturbations f_2 and f_3 be small, say of the order of ε ($f_2 \rightarrow \varepsilon f_2$ and $f_3 \rightarrow \varepsilon f_3$, see, e.g., Eq. (5)). Then, it is natural to assume that, under certain regularity conditions, any solution of the perturbed system is ε -close to the solution of the unperturbed one to which it is expected to converge when $\varepsilon \rightarrow 0$. Accordingly, we assume

$$x_h(t) = x_0(t-s) + \varepsilon x_1(t) + \dots, \quad (10)$$

where $x_1(t)$ no longer contains the arbitrary phase because it is a solution of a non-autonomous system.

Inserting (10) in (9) we get

$$\begin{aligned} d(s) &= \varepsilon \left\{ \int_{-\infty}^{+\infty} f_2[x_0(t-s), \dot{x}_0(t-s)] \dot{x}_0(t-s) dt \right. \\ &\quad \left. + \int_{-\infty}^{+\infty} f_3[x_0(t-s), t] \dot{x}_0(t-s) dt \right\} + \dots \\ &= \varepsilon \left\{ \int_{-\infty}^{+\infty} f_2[x_0(t), \dot{x}_0(t)] \dot{x}_0(t) dt \right. \\ &\quad \left. + \int_{-\infty}^{+\infty} f_3[x_0(t), t+s] \dot{x}_0(t) dt \right\} + \dots, \end{aligned} \quad (11)$$

namely, $d(s) = \varepsilon \{\text{constant} + \text{periodic function of } s\} + \text{higher order } \varepsilon\text{-terms}$. It is important to stress that this expression requires only the knowledge of the unperturbed homoclinic orbit $x_0(t)$, and that the constant part of (11) is due to the damping $f_2(x, \dot{x})$, while the periodic part of (11) is clearly related to the periodicity in time of the excitation $f_3(x, t)$.

It is immediate to check that (11) and (4), which have been obtained under the same hypothesis of nearly Hamiltonian system, not only simultaneously vanish

in the case of homoclinic intersection, but they also have the same mathematical structure and the same functional dependence on the perturbations. It is then possible to conclude that the latter provides the searched energetic justification of the former.

On the basis of the previous considerations we can even obtain formulas for a_0 , $a_1(\omega)$ and $h(\omega s)$, which are the key elements of (4), by a direct comparison. It is easy to check that they provide the *same* expression of the classical Melnikov method [148], so that they are exact in the asymptotic sense.

4.3.7 Minimum Manifolds Distance

Independent of its phenomenological derivation, its physical meaning or its exact calculation by the Melnikov or other methods, we have found that to look for homoclinic intersections we have to search the zeros of the distance (4).

First, we note that, as we are interested in the zeros of (4) and not in its sign, it is possible to assume $a_0 > 0$ without loss of generality. Then, the distance is an oscillating function around a constant positive value, as schematically shown in Fig. 2. From this picture and from (4) it is seen that the distance $d(\omega s)$ has zeros if and only if its minimum

$$\begin{aligned} d &= \min_{m \in [0, 2\pi]} \{d(m)\} = \varepsilon \delta a_0 + \varepsilon \gamma_1 a_1(\omega) \min_{m \in [0, 2\pi]} \{h(m)\} \\ &= \varepsilon \delta a_0 - \varepsilon \gamma_1 a_1(\omega) M \end{aligned} \quad (12)$$

is negative. When d is equal to zero for the first time by varying a governing parameter, for example the excitation amplitude, an homoclinic bifurcation occurs.

Remark. The expression (12) is useful in the case in which a_0 and $a_1(\omega)$ have the same sign. If, on the other hand, they have a different sign, then we must consider

$$\begin{aligned} d &= \min_{m \in [0, 2\pi]} \{d(m)\} = \varepsilon \delta a_0 + \varepsilon \gamma_1 a_1(\omega) \max_{m \in [0, 2\pi]} \{h(m)\} \\ &= \varepsilon \delta a_0 + \varepsilon \gamma_1 a_1(\omega) M \end{aligned} \quad (13)$$

instead of (12).

In the expression (12) a key role is played by the number

$$\begin{aligned} M &= M\left(\frac{\gamma_j}{\gamma_1}, \Psi_j\right) = - \min_{m \in [0, 2\pi]} \{h(m)\} = \\ &= - \min_{m \in [0, 2\pi]} \left\{ \sum_{j=1}^N \left(\frac{\gamma_j}{\gamma_1}\right) \frac{a_1(j\omega)}{a_1(\omega)} \cos(jm + \Psi_j) \right\}. \end{aligned} \quad (14)$$

It is positive, because $h(m)$ has zero mean value, and it does not depend on the overall excitation amplitude $\varepsilon \gamma_1$, which is singled out in (12). Indeed, it is just this property which suggested the previous choice of the overall excitation amplitude.

While being independent of the excitation amplitude, M is instead strongly dependent on the shape of $h(m)$ and, consequently, on the *shape of the excitation*. The most important property of M is that it summarizes in a unique number this dependence, or, equivalently, it summarizes the contribution of the superharmonics added to the reference harmonic excitation (see Eq. (5)).

In the case of the harmonic excitation we have $h(m) = \cos(m)$, so that $M = 1$. This number therefore represents a reference value for M . In fact, we can distinguish between three families of periodic excitations: those for which $M = 1$, $M < 1$ and $M > 1$. The first class is equivalent to the harmonic excitation for what concerns the homoclinic bifurcations. The second class is useful in the problem of control, as we will see in the sequel, while the latter can be used in the case of anti-control [141].

4.4 Influence of the Parameters on the Manifolds Distance

The most important result of the previous section, which is summarized by the expressions (12), is that the minimum distance between the stable and unstable manifolds has been computed as a function of the governing parameters. Roughly speaking, we can distinguish between various families of parameters.

The first family is constituted by the mechanical system parameters such as mass, linear and nonlinear stiffnesses, etc. In (12) they are embedded in the definitions of a_0 and $a_1(\omega)$. For a given mechanical oscillator they are known, while they can be modified in the design process. Their influence cannot be discussed in general, because (i) it strongly depends on the mechanical characteristics of the system (hardening vs. softening, etc.), and (ii) it depends on how a_0 and $a_1(\omega)$ depend on the mass and stiffnesses, and this considerably varies from system to system, as it can be seen by comparing the expressions obtained for different oscillators [37].

The second family consists of the main parameters of the perturbations, such as damping and excitation amplitude and frequency, which are instead singled out in (12). Also these parameters can be considered as data, for a given problem, or can be modified during the design. In the considered case of small perturbations, damping and excitation amplitude influence *linearly* the minimum distance (12), and so their effects can be easily understood. In particular, note that when the damping is zero with non zero excitation, d is always negative, namely, there is always homoclinic intersection in almost Hamiltonian systems (Fig. 1c, d). On the contrary, with damping and without excitation d is always positive so that there is no homoclinic intersection at all (Fig. 1b).

More involved is the influence of the excitation frequency. In this respect, we can only infer that $\lim_{\omega \rightarrow 0} a_1(\omega) = 0$, because $\omega = 0$ corresponds to static excitation which does not pump energy into the system, and that $\lim_{\omega \rightarrow \infty} a_1(\omega) = 0$, because when the frequency is very high, only the time average of the excitation actually forces any given inertial system: but since the excitation is periodic, the average

is zero, so that the system is basically not forced, and the oscillating term in the distance disappears.

The last family we consider is constituted by the excitation shape parameters γ_j/γ_1 and Ψ_j , which are the key ingredients for the forthcoming analysis. Their influence on the minimum distance is summarized by the number M , which basically plays the role of an interface between the excitation shape and d . Not only are these parameters very important in the control method which is the object of this section (because they are allowed to vary for control design purposes), but they are also the less common because usually the excitation is considered as harmonic (sinusoidal). Thus, they deserve special attention.

Actually, the dependence of d on M is linear, and thus it is trivial. The non trivial point is instead the (nonlinear) dependence of M on γ_j/γ_1 and Ψ_j . The things are quite involved because the dependence of M on γ_j/γ_1 is modulated by the factors $\frac{a_1(j\omega)}{a_1(\omega)}$ appearing in the definition of h_j . Since these factors are system dependent, because so is $a_1(s)$, we cannot discuss in general the dependence of M on γ_j/γ_1 .

What we can do in a *unified* way, i.e. without referring to a specific mechanical system, is to analyze the dependence of M on h_j (and on Ψ_j). To illustrate this fact by an example, we consider $N = 2$, i.e. a single added superharmonic. The function $M(h_2, \Psi_2) = -\min_{m \in [0, 2\pi]} \{\cos(m) + h_2 \cos(2m + \Psi_2)\}$ is plotted in Fig. 3, from which we see that both h_2 and Ψ_2 strongly influence M . More precisely, we clearly see that there are regions for which $M < 1$, which will be useful for control, and regions for which $M > 1$, which are useless (dangerous indeed) with respect to the control features. These conclusions are general and hold also for the case $N > 2$.

4.5 Homoclinic Bifurcation Thresholds

The multidimensional parameters space can be divided in two regions, according to whether the manifolds minimum distance is positive, corresponding to non intersecting, or detached, manifolds, or the minimum distance is negative, corresponding to intersection. The boundary between these two regions, i.e., the locus of points $d = 0$, corresponds to manifolds tangency, namely, to the homoclinic, or heteroclinic, bifurcation. In fact, crossing this boundary the manifolds become intersecting (or detach, according to the direction of crossing).

The detection of this boundary is crucial for the control method based on the elimination of the homoclinic or heteroclinic bifurcations. In fact, the method can also be interpreted as aimed at shifting this boundary in a certain direction, usually toward higher excitation amplitudes.

It is common to represent the bifurcation threshold in terms of excitation amplitude, namely, to solve $d = 0$ with respect to γ_1 . As a matter of fact, this is one of the reasons why we pay a lot of attention to the definition of a proper excitation amplitude along this section.

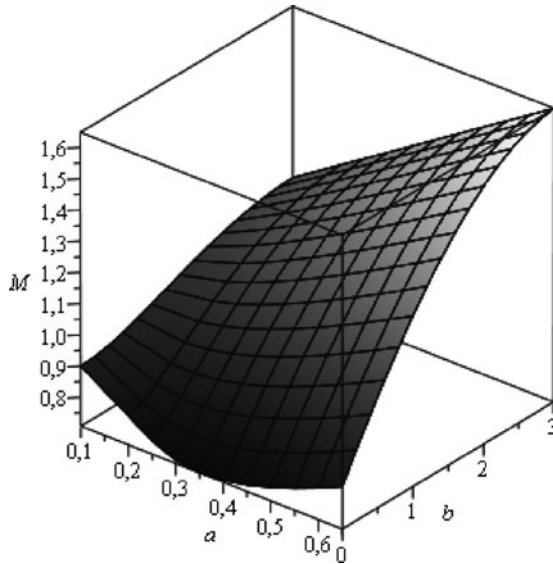


Fig. 3 The function $M(a, b)$, where $a = h_2 = \left(\frac{\gamma_2}{\gamma_1}\right) \frac{a_1(2\omega)}{a_1(\omega)} \in [0, 0.65]$ and $b = \psi_2 \in [0, \pi]$

The solution of $d = 0$ represents the critical excitation amplitude threshold for which the homoclinic or heteroclinic bifurcation takes place, and it will be denoted by $\gamma_{1,cr}$. If the distance can be expressed by (12) the computation of the critical threshold is trivial:

$$\gamma_{1,cr} = \frac{\delta a_0}{a_1(\omega)} \frac{1}{M}. \quad (15)$$

The simplicity of the previous expression is obviously a consequence of the linear nature of the distance with respect to γ_1 , which however no longer holds in more general cases, such as, for example, those in which the manifolds distance is computed numerically [136]. Here, the solution of $d = 0$, while being conceptually simple, requires an extra computational effort consisting of solving a nonlinear algebraic equation. Furthermore, the equation is known only in a discrete set of points, and this is another element of difficulty.

In any case, irrespective of how the critical excitation amplitude has been computed, exactly, like in piece-wise linear systems [62, 140], by a perturbative approach, like in (15), or numerically, this value is eventually a function of all the other parameters of the system.

It is interesting that the excitation shape enters the expression (15) only by means of the single number M , which evidently summarizes also in this respect the effects of the superharmonics added to the basic harmonic excitation. Of course, this property is not expected to hold in general, and in other cases the dependence of $\gamma_{1,cr}$ upon the excitation shape is more involved.

We remark that both the minimum distance d (through Eq. (12)) and the homoclinic bifurcation threshold $\gamma_{1,\text{cr}}$ (through Eq. (15)) are inversely proportional to the excitation shape parameter M , i.e., the smaller M , the larger d and $\gamma_{1,\text{cr}}$. The influence of the other parameters is complex, and in general it is far from linear. In fact, only the damping δ enters linearly in (15), this being a consequence of its smallness.

In general, not even a qualitative estimate is possible a priori. For example, for some mechanical systems the critical threshold of the homoclinic bifurcation of the hilltop saddle tends to a finite value when the excitation frequency tends to zero (e.g., the inverted pendulum between lateral barriers [62]) while in others it is unbounded (e.g., the Helmholtz oscillator [137]). This is a consequence, or perhaps a source, of the fact that chaotic behaviour – which is strongly related to global bifurcations – is not monotonic with respect to any parameter.

A special situation occurs for the class of excitations having $M = 1$, which includes the harmonic excitation. Since it is considered as the reference one, we indicate by $\gamma_{1,\text{cr}}^h = \frac{\delta a_0}{a_1(\omega)}$ the corresponding critical threshold. According to this definition, Eq. (15) can be written in the equivalent form

$$\gamma_{1,\text{cr}} = \gamma_{1,\text{cr}}^h \frac{1}{M}. \quad (16)$$

This expression has the advantage of splitting into two independent parts the main dependences of $\gamma_{1,\text{cr}}$ upon the system parameters. In fact, $\gamma_{1,\text{cr}}^h$ contains all information regarding the mechanical system, such as mass, stiffness, etc., and the *reference* excitation, such as its kind (i.e., external, parametric, etc.) and frequency, while M contains information only on the excitation shape (or on the *controlling* part of the excitation).

This property does not hold in more general situations, for example when the critical threshold is computed by numerical methods without any assumptions on the smallness of the excitation amplitude. Here all the parameters of the systems are strongly coupled, and it is not possible to single out the dependence on the excitation shape.

When the excitation is periodic, irrespective of being harmonic (i.e., sinusoidal) or not, external or parametric, etc., it is usual to draw the critical amplitude as a function of the excitation frequency, for fixed values of all the other parameters. An example referring to the Helmholtz oscillator [137] is reported in Fig. 4. Here the excitation is periodic, and accordingly, the critical threshold is $\gamma_{1,\text{cr}}^h$.

From Fig. 4 it is clear how the parameters subspace is divided in the two regions of homoclinic intersection and homoclinic detachment. The first one is the region of the so-called “Melnikov chaos,” where various manifestations of chaos are expected: some of them, such as fractal basin boundaries, sensitivity to initial conditions, etc., will certainly occur, although to a different extent, while others, such as the chaotic attractors, are only theoretically feasible, but not necessarily exist in the whole region. When chaos – in all its manifestations – is considered as an undesirable

event, this region must be regarded as unfeasible, or dangerous, and care is needed in staying sufficiently far away from it.

The region of Melnikov chaos is “unbounded” and convex in Fig. 4. There are no reasons to believe that these properties are general, and both bounded and/or non convex sets of homoclinic intersection are expected to occur, at least in principle. Even non-connected sets are possible.

4.6 Control Ideas

So far we have obtained all the necessary prerequisites for application of the control method, i.e., we have related the manifolds minimum distance d and the homoclinic (or heteroclinic) bifurcation threshold $\gamma_{l,cr}$ to the system parameters, and we are now in a position to describe the control ideas.

There are two conceptually different approaches, which however in some cases are equivalent and yield the same control implementation.

In a *fixed* situation, namely, for a given system, given operational conditions and, therefore, given system parameters, one can be interested in eliminating the stable and unstable manifolds intersection by the action of control. This entails acting on the system in such a way that the minimum manifolds distance passes from negative to positive. Global bifurcations are not directly involved in this case, although assuming that there is a continuous passage from the uncontrolled (intersection, negative distance) to the controlled (no intersection, positive distance) system, obvi-

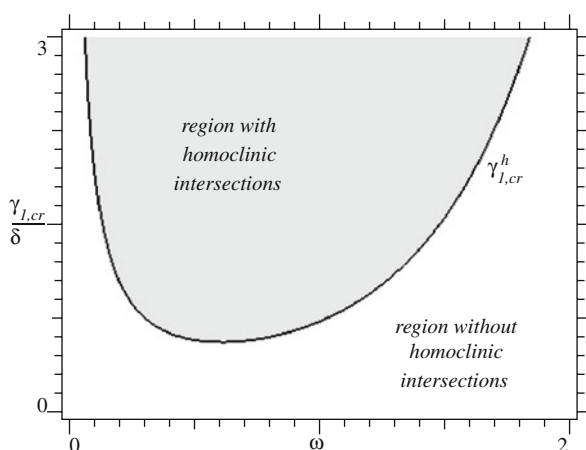


Fig. 4 The critical threshold $\gamma_{l,cr}^h$ as a function of the excitation frequency ω , for the Helmholtz oscillator [137]

ously there is an intermediate condition in which the distance is zero, corresponding to a homoclinic (or heteroclinic) bifurcation.

This approach can be of interest in specific situations, and when an overall picture of the control properties is not needed. Accordingly, only a “local” analysis is required.

For *varying* system parameters (e.g., the excitation amplitude), on the other hand, one can be interested in shifting the homoclinic (or heteroclinic) bifurcation threshold in parameters space by the control action. Herein, the global bifurcation becomes the key point, and in principle the analysis is more complete and, consequently, more difficult. Practically this approach is aimed at modifying the regions of non homoclinic bifurcation of Fig. 4 by changing their boundaries.

By comparing the two approaches, one notes that the former is the most natural when the control is considered from a dynamical system point of view, as it involves only topological properties, while the latter is the most natural when the control is considered from an engineering point of view, because it concerns also practical aspects, such as the behaviour in parameters space, which have important consequences in terms of system response.

Actually, if the two control targets (manifolds distance and critical excitation amplitude) have the same dependence on problem parameters, the two approaches are practically equivalent, though remaining conceptually distinct. An important example is that in which we consider the homoclinic bifurcations of nearly Hamiltonian systems, detected by the Melnikov method (see previous sections), and assume the shape of the excitation as control action. We have shown in (12) and (14) that the minimum distance and the homoclinic bifurcation threshold, respectively, are both inversely proportional to the shape parameter M , so that in both cases *control entails decreasing M* .

In more general cases, such as those in which the distance is computed numerically without any specific restriction on the system, the two approaches remain distinct also from a practical point of view, and in principle they lead to practically different results, although similar qualitative behaviours are expected in common situations.

4.7 Gains and Saved Region

To quantify the improvement of control with respect to the reference (harmonic) excitation in terms of homoclinic bifurcation threshold, it is useful to introduce the so-called *gain*

$$G = \frac{\gamma_{1,\text{cr}}}{\gamma_{1,\text{cr}}^h} = \frac{1}{M}, \quad (17)$$

which is the ratio between the critical threshold with control and the critical threshold in the reference case.

With this definition, we can reformulate the control idea by saying that it consists in increasing the gain by varying γ_j and Ψ_j .

The region $\gamma_{1,cr}^h(\omega) < \gamma_1 < \gamma_{1,cr}(\omega) = G\gamma_{1,cr}^h(\omega)$ is the zone of parameters space where arbitrary periodic excitations are theoretically effective in eliminating homoclinic chaos (with respect to the harmonic excitation), and it has been called *saved*, or *controlled*, region [37]. Thus, another, equivalent, point of view about the control method is that it is aimed at obtaining a saved region in the excitation amplitude-frequency parameters space.

The saved region for Helmholtz systems, whose homoclinic bifurcation threshold is depicted in Fig. 4, is reported in Fig. 5 in the case of an ω -independent value of the gain G . From this picture it is evident how the saved region is obtained by shifting the homoclinic bifurcation threshold toward higher excitation amplitudes.

4.8 Optimal Control and Optimization Problems

In the previous sections it has been established that the control method consists in varying the shape of the excitation in order (i) to increase the distance between stable and unstable manifolds, or (ii) to increase the homoclinic bifurcation threshold, or (iii) to obtain a saved region. It is natural to do these operations in a clever way, i.e., by choosing among all admissible excitations the one which *maximizes* the considered target function (d , $\gamma_{1,cr}$ or the magnitude of the saved region). This point represents the second keystone of the considered control method, and is discussed in this section.

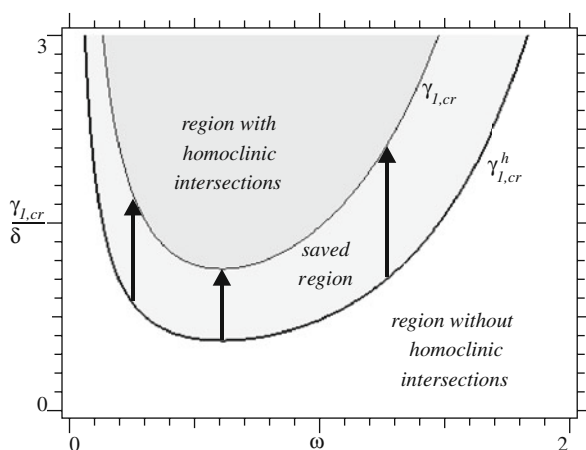


Fig. 5 The saved region for the Helmholtz oscillator [137] of Fig. 4

In both cases (i) and (ii), pursuing the objective to maximize the manifolds distance or the homoclinic critical threshold entails increasing the gain G introduced in (17). But choosing among all admissible excitations the one which maximizes the gain leads to the optimization problem

Problem P0. Maximize G by varying the Fourier coefficients γ_j and Ψ_j of the excitation,

whose solution provides the best or *optimal* excitation. Since increasing G entails decreasing $M = 1/G$, and decreasing the positive number M entails increasing

$$-M = \min_{m \in [0, 2\pi]} \{h(m)\},$$

it is possible to rewrite problem $P0$ in its extended form:

Problem P0'. Maximize $\min_{m \in [0, 2\pi]} \{h(m)\}$ by varying the coefficients γ_j and Ψ_j of $h(m) = \sum_{j=1}^N \left(\frac{\gamma_j}{\gamma_1} \right) \frac{a_1(j\omega)}{a_1(\omega)} \cos(jm + \Psi_j)$.

4.8.1 Universal Optimization Problem

It is useful to reformulate problem $P0$ in the following equivalent form:

Problem P1. Maximize $\min_{m \in [0, 2\pi]} \{h(m)\}$ by varying the Fourier coefficients h_j and Ψ_j of $h(m) = \cos(m) + \sum_{j=2}^N h_j \cos(jm + \Psi_j)$.

The key advantage in passing from problem $P0$ to problem $P1$ is that the latter is *system-independent*, i.e. *universal*. This can be ascertain by noticing that in $P0$ the system-dependent function $a_1(\omega)$ is present, while it disappears in $P1$, where the variable coefficients γ_j are “replaced” by the variable system-independent coefficients h_j .

Note that the function $a_1(\omega)$ not only is system dependent, but in general it also depends on the specific homoclinic orbit considered in systems having more than one homoclinic intersection.

Being universal, problem $P1$ can be solved just once, and its solution is universally valid, namely, it is *not* necessary to solve problem $P1$ for each specific system.

To illustrate the solution of problem $P1$, let us first consider the simplest case $N = 2$, i.e., the excitation is constituted by the basic harmonic plus one controlling superharmonic. This case can be handled manually [137]. In fact, the optimal choice is $\Psi_2 = 0$, and

$$\begin{aligned} M(h_2) &= - \min_{m \in [0, 2\pi]} \{h(m)\} \\ &= - \min_{m \in [0, 2\pi]} \{\cos(m) + h_2 \cos(2m)\} \\ &= \min \left\{ 1 - h_2, h_2 + \frac{1}{8h_2} \right\}, \end{aligned} \quad (18)$$

so that $G(h_2) = 1/M(h_2)$. This function is depicted in Fig. 6, from which we infer that

$$G_{\text{optimal}}^{N=2} = \max_{h_2} \{G(h_2)\} = \sqrt{2}, \quad (19)$$

this maximum being attained for $h_2 = \sqrt{2}/4$.

The cases with more superharmonics are conceptually similar, but computationally more difficult, and they have been solved numerically after having proved that the condition $\Psi_j = 0$ is still optimal [37]. The results are reported in Table 1, which also shows that for $N \rightarrow \infty$ the solution $h(m)$ tends to a well defined limit constituted by a positive Dirac delta of amplitude π at $m = 0$ plus the constant function $-1/2$ (whose Fourier coefficients are in fact $h_j = 1$ and $\Psi_j = 0$). It is proved in [26, 62] that this last function actually represents the mathematical solution of the optimization problem in the case of an infinite number of superharmonics.

Table 1 The numerical results of various optimization problems with increasing number of superharmonics

N	G_{optimal}^N	h_2	h_3	h_4	h_5
2	1.4142	0.353553			
3	1.6180	0.552756	0.170789		
4	1.7321	0.673525	0.333274	0.096175	
5	1.8019	0.751654	0.462136	0.215156	0.059632
...	...	...	...	...	...
∞	2	1	1	1	1

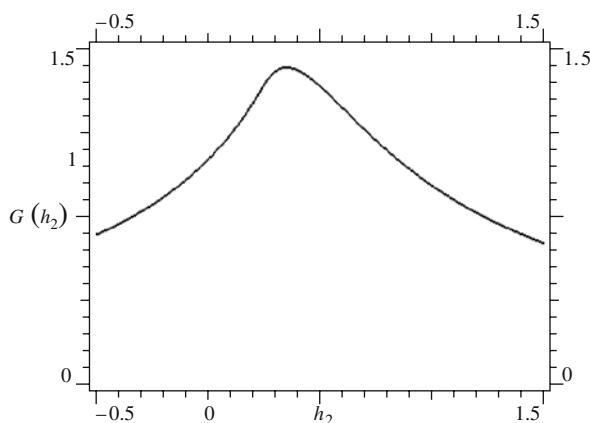


Fig. 6 The function $G(h_2)$ in the case of a single superharmonic ($N = 2$)

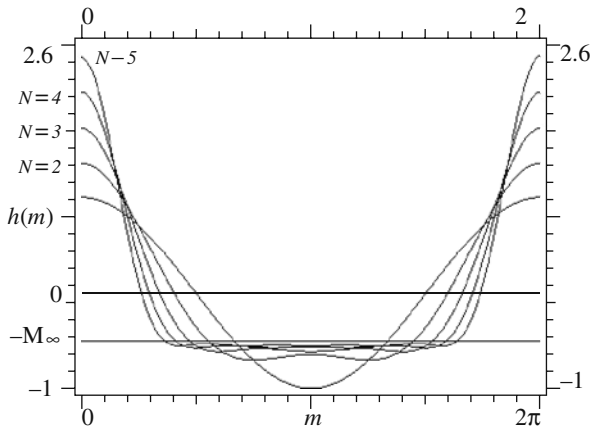


Fig. 7 Different optimal functions $h(m)$ for $N = 2, \dots, 5$

Examples of optimal $h(m)$ are reported in Fig. 7 from which it is clear that the optimal solution is characterized by the presence of multiple coinciding minima. It is also possible to foresee the convergence of that family of solutions to the limit solution for $N \rightarrow \infty$ discussed above.

4.8.2 From the Universal Optimal Solution to the Real Optimal Excitation

The link between the optimal solutions obtained in the previous point and the real optimal excitation to be applied to the structure is represented by the relations

$$\gamma_j = h_j \gamma_1 \frac{a_1(\omega)}{a_1(j\omega)}, \quad (20)$$

so that the optimal excitation can be written as

$$f_3(x, t) = \varepsilon \gamma_1 a_1(\omega) \sum_{j=1}^N \frac{h_j}{a_1(j\omega)} \sin(j\omega t), \quad (21)$$

which, when compared with (5) provides the optimal controlling excitation

$$\bar{f}_3(x, t) = a_1(\omega) \sum_{j=2}^N \frac{h_j}{a_1(j\omega)} \sin(j\omega t). \quad (22)$$

In (21) and (22) there are the “universal” coefficients h_j , which can be taken from Table 1 and do *not* need to be recalculated each time, and the function $a_1(\omega)$ which is

instead system dependent and thus changes when the control is applied to different cases.

The previous consideration further underlines one interesting, and useful, aspect of the proposed control method. In fact, upon solving the difficult part, namely the solution of the mathematical optimization problem, the operator is just left with the “technical” problem of computing the function $a_1(\omega)$ which can be “easily” done by looking at the manifolds behaviour, as discussed in the previous sub-sections. This makes the application of the control method straightforward.

Another issue arises when passing from the universal solution to practical applications. In fact, for technical reasons it may be useful to have some limitations on the optimal solution (for example, one can ask that $|\gamma_f| < |\gamma_1|$, or that $\max_{t \in [0, T]} \{\tilde{f}_3(x, t)\} < \text{const.}$, etc.). These requirements can be introduced as *constraints* in the optimization problem *PI*. While not introducing special conceptual difficulties (although more sophisticated tools may be required to search for the optimal solution), these constraints are usually system dependent, so that the optimization problem turns into a *non universal* one, which is not generic and has to be solved separately for each system. This makes the application of the control method much more involved. Constrained optimization problems have been solved in [137].

4.9 Extended (“Global”) and Localized (“One-Side”) Application of Control

In many mechanical systems, such as symmetric oscillators, there is more than one homoclinic bifurcation which is relevant for complex dynamics and needs to be controlled (i.e., eliminated or shifted in parameter space).

In this situation, the presence of two or more homoclinic orbits permits a choice between different control strategies. Indeed, we can:

1. control *only* one (e.g., the right *or* the left in symmetric systems) homoclinic bifurcation, irrespective of what happens to the others,
2. control *simultaneously* two or more of them (e.g., the right *and* the left in symmetric systems),
3. control *simultaneously* all of them.

The choice 1 (called “one-side control” [37]) is aimed at obtaining a topologically localized control, and can be pursued by applying the points of the previous section separately to each homoclinic loop. Of course, in this case we will have an optimal excitation for each homoclinic intersection to be removed, and since these are system- and orbit-dependent, there is no hope that the optimal excitation able to remove one homoclinic intersection also succeeds in removing the other homoclinic intersections (indeed, the non-controlled homoclinic bifurcation threshold is usually lowered).

The choice 2 is aimed at controlling an extended part of the phase space, while the choice 3 is aimed at controlling the whole phase space, and thus it has been

called “global control” [37]. They require further considerations, as a consequence of the fact that in this case we want to control *simultaneously* more homoclinic intersections, i.e., we have to spread the resources available for optimization over different undesired events.

The first characteristic property of this case is that the underlying optimization problem is *no longer universal*, apart from the special case of symmetric systems with two symmetric homoclinic loops. Unfortunately, things are even more complex, and different approaches/strategies are possible. A detailed description of the whole scenario is reported in [139] with reference to the Helmholtz – Duffing oscillator. Some main results are summarized in the following.

4.9.1 Global Control of Gains

Aiming at controlling simultaneously more homoclinic bifurcations, we have to monitor the control target (distance d or homoclinic bifurcation threshold $\gamma_{1,cr}$) of each homoclinic loop. From a mathematical point of view this goal can be obtained by controlling the minimum value among all bifurcations, which guarantees that all critical thresholds will not be reduced.

Since both the distance and the homoclinic bifurcation thresholds are proportional to the gain G , the first possible approach consists of maximizing the minimum gain, which yields the following optimization problem:

Problem P2. Maximize $G = \min\{G^1, G^2, \dots, G^L\}$ by varying the Fourier coefficients γ_j and Ψ_j of the excitation.

In the problem *P2* we have that $G^k = 1/M^k$ and

$$M^k = - \min_{m \in [0, 2\pi]} \left\{ \sum_{j=1}^N \left(\frac{\gamma_j}{\gamma_1} \right) \frac{a_1^k(j\omega)}{a_1^k(\omega)} \cos(jm + \Psi_j) \right\}, \quad (23)$$

where $k = 1, 2, \dots, L$ is the number of homoclinic bifurcations to be controlled simultaneously, and $a_1^k(\omega)$ is the function corresponding to the k -th homoclinic loop. Note that in general all M^k s are distinct from each other.

The (unavoidable) presence of the functions $a_1^k(\omega)$ in problem *P2* shows that it is system-dependent.

In [139], where only the case $L = 2$ is considered, this problem has been named “global control *without* symmetrization,” because the homoclinic bifurcation thresholds $\gamma_{1,cr}^k$, which are distinct in the reference case, remain distinct also in the optimal case. In the same paper the optimization problem has been solved in some specific cases, showing that

1. the optimal gain is much lower than that obtained in the “one-side control” (see Table 1);
2. at the optimum, the condition $G^1 = G^2$ is attained.

Roughly speaking, property 1 is a consequence of the fact that now the control “efforts” are spread over more homoclinic orbits, instead of being concentrated on a single one. This property is expected to be general, contrary to property 2 which is expected not to be satisfied, especially in the presence of more homoclinic loops.

4.9.2 Global Control of Homoclinic Bifurcation Thresholds

In the previous strategy the homoclinic bifurcation thresholds remain distinct after the application of control. It could be desirable to have a control providing, or attempting to provide, the same homoclinic bifurcation thresholds, even at the price of reducing some of them with respect to the values for harmonic excitation. This consideration leads to another optimization problem:

Problem P3. Maximize $\gamma_{1,\text{cr}} = \min \left\{ \gamma_{1,\text{cr}}^1, \gamma_{1,\text{cr}}^2, \dots, \gamma_{1,\text{cr}}^L \right\}$ by varying the Fourier coefficients γ_j and Ψ_j of the excitation,

where

$$\gamma_{1,\text{cr}}^k = \frac{\delta a_0^k}{a_1^k(\omega)} \frac{1}{M^k}. \quad (24)$$

Note that, contrary to the previous case, now also the different expressions a_0^k are taken into account – and this further stresses how also this case is system-dependent. In [139] this problem has been called “global control *with* symmetrization.”

For increasing control “resources,” the optimization problem works in the following way. It increases the lowest value of $\gamma_{1,\text{cr}}^k$, say $\gamma_{1,\text{cr}}^3$, until it reaches the subsequent lowest value (which could be lowered – with respect to the harmonic case – during this first phase), say $\gamma_{1,\text{cr}}^5$. Then, it increases simultaneously $\gamma_{1,\text{cr}}^3$ and $\gamma_{1,\text{cr}}^5$, until they reach the subsequent lowest value (again, possibly decreased), and so on. This process ends when:

1. The “resources” for control finish *after* having equalized all the $\gamma_{1,\text{cr}}^k$. This case has been named “global control with *achieved* symmetrization” in [139], and the optimal solution is characterized by $\gamma_{1,\text{cr}}^1 = \gamma_{1,\text{cr}}^2 = \dots = \gamma_{1,\text{cr}}^L$, which represents a desired circumstance.
2. The “resources” for control finish *before* having equalized all the $\gamma_{1,\text{cr}}^k$. This case has been named “global control with *pursued* symmetrization” in [139].

Of course, the former case is expected when the homoclinic bifurcations of the different loops under harmonic excitation are close to each other, as occurs, for example, for weakly asymmetric systems.

The latter case, on the other hand, is expected when the reference homoclinic bifurcations are very different from each other; a limit situation is represented by the case in which the reference homoclinic bifurcation thresholds are so different that control uses all its resources to increment the lowest value, so that in this case this approach coincides with the “one-side control.”

4.10 On the Application of the Control Methods to Archetypal Single-d.o.f. Systems

In the previous sub-sections, a phenomenological analysis of the homoclinic and heteroclinic bifurcations of the invariant manifolds of a saddle triggering the occurrence of complex events in dynamical systems has been presented for a generic, periodically excited, oscillator, along with the basic idea and the objective of their control, and some ensuing optimization problems.

The next issue consists in applying the control method to specific nonlinear oscillators. This has been the subject of a number of research papers [62, 63, 107, 135–146] which the reader is referred to for the details of successful application. Herein, we just aim at giving an overview of some main characterizing features of the considered systems. In this respect, it is important to refer to a suitable number of archetypal oscillators with a well-defined mathematical character and such to reliably describe the physical behaviour of real systems in *engineering science* and *mechanics*. Of course, their presentation and discussion has to be suitably organized. Indeed, nonlinear oscillators can be classified based on different viewpoints, according to whether the focus is preferably on dynamical or control aspects, while paying attention – in any case – to the characterizing features of physical world applications and to the associated computational aspects.

Herein, with the aim of stressing the mechanics environment wherein this chapter – devoted to exploiting system dynamics for control purposes – has nurtured, we choose to assume a basically *mechanical* viewpoint with the associated *dynamic* phenomena in the background, by highlighting those features of the considered systems which entail more meaningful *control* aspects. Within this overall scheme, systems are categorized according to different features, namely whether they are (i) smooth or non-smooth, whether they are characterized by (ii) single-well or multi-well potentials, by (iii) softening or hardening behaviour, by (iv) symmetric or asymmetric characteristics, also considering the occurrence or non-occurrence of a system natural frequency, and whether the technical interest is towards their (v) transient or steady response. Of course the necessary attention is also paid to the characterizing mechanical phenomena to be possibly controlled.

In terms of formulation and implementation of the dynamically based control strategy, a second viewpoint pays due attention to those *geometrical* and *computational* aspects which play a fundamental role in describing and understanding the system dynamics and in evaluating the performance of the control procedure, respectively. In this respect, the most meaningful alternatives are concerned, on one side, with dealing with either (i) homoclinic or heteroclinic manifolds, and with controlling either (ii) overall or local bifurcational events; on the other side, with the possibility to develop an (i) approximate or exact bifurcation analysis, and to implement a (ii) mainly theoretical or purely numerical control strategy. Of course, strong cross-correlations do occur amongst several of the above mentioned classification criteria. They are however believed to maintain their own interest in the aim of schematically organizing a quite rich and involved scenario of mechanical systems as well as of comparing the respective dynamic behaviours.

4.10.1 Smooth vs. Non-smooth Systems

Smoothness or non-smoothness are a basic property of mechanical systems with important consequences as regards their mathematical description and dynamical characterization. Thus, they are assumed as a first criterion to distinguish between two broad classes of systems.

Due to the occurrence of discontinuities in either the displacement or the velocity, non-smooth systems exhibit a further enriched pattern of response classes and local/global bifurcations with respect to the already involved scenario of smooth systems, along with some main novelties in the dynamics features.

In terms of the proposed control, the main difference is that in smooth systems the homoclinic bifurcation cannot be computed analytically, not even for the homoclinic orbits of hilltop saddles. The only analytical results are obtained by perturbative approaches, as the Melnikov method [148] or the method illustrated in Sect. 4.3. For the piece-wise linear systems, instead, the homoclinic bifurcation threshold can be computed exactly, and this allows a more detailed application of control.

A number of fully nonlinear oscillators with continuous-type nonlinearities, of either exact (such as the mathematical pendulum) or approximate, polynomial-kind (such as the Duffing [136, 138], Helmholtz [137], Helmholtz-Duffing [139] oscillators, a micromechanical system [142]) nature, have been considered. The latter mostly – though not exclusively – describe the nonlinear single-mode, non-internally resonant, dynamics of a number of smooth infinite-dimensional systems of interest in structural and engineering mechanics.

Two piecewise linear oscillators have been dealt with, i.e. the impacting inverted pendulum [26, 62, 135, 107] and the rocking rigid block [63, 140, 141], to be intended, respectively, as paradigmatic models for non-smooth nonlinear dynamics of mathematical/mechanical significance or more practical interest.

4.10.2 Single-Well vs. Multi-Well Potentials

Depending on the energy imparted to the system through the initial conditions or the excitation amplitude, as well as on the values of other control parameters, two main issues are of considerable practical importance for a mechanical system, namely (i) the possibility to have the system actually working within the range of operating conditions which it has been designed for, and (ii) the loss of technical performance or structural integrity ensuing from the system getting off its foreseen operational regime.

In dynamical terms, it is possible to restrict ourselves to systems characterized by the existence of either a single-well or a (possibly) multi-well potential, for which two different critical situations may occur, namely,

1. the motion does develop on a *small scale* in the phase space, being actually restricted within a sole, i.e. the foreseen, potential well, however with dynamical features other than the wanted ones, as it may occur due to any in-well bifurcation of the system dynamics to a different periodic regime or to an aperiodic regime;

2. the motion does develop on a *large scale* in the phase space, i.e. also beyond the invariant manifolds of the hilltop saddle which delimit the working potential well and organize the whole system dynamics.

The occurrence of either a single-well or a multiple well has a considerable influence as regards various dynamics, bifurcation, and control issues.

4.10.3 Softening vs. Hardening Systems

The distinction between hardening and softening systems reflects the intrinsic mathematical/mechanical character of the system constitutive law and the associated dynamic effects occurring for finite amplitude motions. It mostly pertains to the large scale motions arising upon the global bifurcation of the hilltop saddle manifolds, which can occur in both single-well and multiple-well systems, and it is actually concerned with the kind of out-of-well attractor onto which the system settles down upon overcoming the bounding invariant manifolds, i.e. after escaping from the potential well. Again, two basically different situations can occur.

1. Though being no more restricted within the reference well, the overall system dynamics still remains *bounded*. This is the case where, after escape, the motion develops entirely within a neighbouring, bounded, well, or it wanders around, e.g., two wells, the reference and a neighbouring one. Scenarios of this kind may occur in multi-well hardening systems and are usually unpleasant from the application viewpoint, yet not destructive. Aperiodic motion scattered between adjacent wells, or the so called cross-well chaos, are indeed classes of motion pertaining to such after-escape non-destructive regimes, and are quite common in several (e.g., two-well Duffing-type [136, 138, 139]) systems. Of course, such kinds of unwanted dynamics may need to be avoided or somehow controlled. However, technical situations can be devised wherein realizing alternating dynamic regimes between the reference well and an adjacent one can even be desirable: let us think, e.g., of the shift between oscillating (in-well) and rotating (out-of-well) regimes in a planar pendulum subjected to vertical harmonic (i.e. parametric) excitation of its support [150].
2. The system dynamics escapes to “infinity”. This is the case where, after escape, the motion is theoretically *unbounded*, and practically corresponds to the system settling down onto an attractor far away and completely different from the designed one or, in other terms, to the definitive system failure. Mechanical situations of reference in this respect are, e.g., the capsizing of a rolling ship in ocean waves [7, 137, 151], the overturning of a rocking rigid block under periodic acceleration of its basement [140, 141], or the pull-in phenomenon occurring in a microelectromechanical system [142] to be used as a resonator or sensor device, which corresponds to the microbeam collapse onto the electrically charged substrate. However, there are again technical situations wherein realizing a regime nominally “at infinity” is highly desirable: think, e.g., of the pull-in dynamics of the microbeam in a MEMS to be used as a switch device.

It is also worth stressing two more meaningful points. (a) Both the two different escape situations just illustrated are actually characterized, from a topological viewpoint, by a progressive erosion process of either a single in-well basin of attraction or of the whole collection of in-well basins. Yet, the degree of loss of dynamical integrity [146, 147] and the criticality of the final outcomes are considerably different in the two cases. (b) The scenario of bounded/unbounded dynamical regimes of a given system in the excitation parameters space can be influenced by the occurrence or non-occurrence of a system natural frequency, the former case naturally entailing a possibility of resonant excitation condition. This may meaningfully affect the (major or minor) system sensitivity, in terms of loss of dynamical integrity, to the variation of a leading bifurcation parameter.

4.10.4 Homoclinic vs. Heteroclinic Bifurcations

As already discussed, the occurrence of a generic underlying bifurcational event, i.e. of the global, homoclinic or heteroclinic, bifurcation of a (possibly) hilltop saddle governing the whole system dynamics (or most of it), is the necessary prerequisite for developing a generic approach to control – or design – of mechanical systems. Yet, several distinct topological situations do occur for different oscillators. By way of example, in the phase portrait one can have (i) one homoclinic orbit surrounding the unique potential well, like in Helmholtz-type softening oscillators [137], (ii) two homoclinic orbits surrounding two separate wells, joined through the intermediate hilltop saddle, like in Duffing-type hardening oscillators [136, 138, 139] – in both cases irrespective of the nature of the system being smooth or non-smooth –, or (iii) one couple of heteroclinic orbits, each one connecting the two organizing hilltop saddles, which surround the unique potential well, like in the Duffing-type softening oscillators or in the rocking block [63, 140, 141]. Of course, more involved phase portraits, characterized by the occurrence of both homoclinic and heteroclinic orbits, may also occur.

While in the case of a single (homoclinic) unperturbed manifold there is only one possible global bifurcation corresponding to the tangency of the split perturbed manifolds, in the case of two distinct (either homoclinic or heteroclinic) unperturbed manifolds the global bifurcations can occur either simultaneously or separately, depending on the values of some varying perturbation parameter. And this has very important consequences in terms of control, as we have seen above.

It is worth mentioning how, in a given system, the homo/heteroclinic bifurcation of the invariant manifolds of an hilltop saddle – i.e. of a globally organizing saddle – does represent the main bifurcational event governing the transition of system dynamics from a small scale to a large scale regime. Yet, as a rule, other local saddles (up to possibly infinitely many) also exist within each potential well, as a consequence of either the instability of a single in-well periodic solution or of the onset of a chaotic saddle born at a previous homo/heteroclinic bifurcation of the coexisting hilltop. Though being *secondary* bifurcational events, the intersections of the invariant manifolds of a local saddle may also entail meaningful changes of system dynamics, either on the small scale (i.e., in-well) regime or on the large

scale (i.e., cross-well) regime. The first one may correspond, e.g., to the transition from a locally periodic to a locally complex (in-well chaos) response. The second, usually as a consequence of the involvement in the global bifurcational event of also the hilltop saddle due to some heteroclinic connection, may correspond, e.g., to the transition from in-well to cross-well chaos which occurs according to any “turning around” topological mechanism (see, e.g., [152]).

Before addressing the control issue in various possible situations, two more *dynamical* issues, which are also representative of major mechanical differences, are mentioned.

4.10.5 Symmetric vs. Asymmetric Systems

The symmetry or asymmetry in the force-displacement relationship of the underlying mechanical system do reflect themselves in distinguishing features in the potential and the unperturbed phase space of the associated nonlinear oscillators.

This distinction deserves attention mostly from a dynamical systems point of view. Focusing on the more “dangerous” softening oscillators, let us think of the difference – under symmetric excitation – between a single-well potential with one escape direction (the smooth Helmholtz oscillator or the non-smooth monolaterally unconstrained rocking block) or with two asymmetric escape directions (the Helmholtz-Duffing oscillator), on one side, and a single-well potential with two symmetric escape directions (the Duffing oscillator or the bilaterally unconstrained rocking block), on the other side. The solely or prevailing sensitivity to escape ensuing from the asymmetry feature, respectively associated with the occurrence of either one single hilltop saddle or of two hilltop saddles of different energy level, does correspond, in the various cases, to weaker, asymmetric, mechanical features in the background.

In contrast, the symmetry entails, e.g., in softening systems – such as the Duffing or the rigid block oscillators – that the potential well is surrounded by a couple of heteroclinic orbits which become a unique heteroclinic loop connecting two equal energy hilltop saddles in the presence of imperfections, whereas in hardening systems – such as the two-well Duffing oscillator – the two homoclinic bifurcations occur simultaneously for all symmetric excitations.

Moreover, still thinking in dynamical terms, it must be underlined how, apart from some pathological situations, symmetric oscillators are structurally unstable and can be considered as particular cases of asymmetric oscillators, though they are not always a limit, in an appropriate sense, of the latter (see, e.g., the softening Helmholtz-Duffing equation in [137]).

4.10.6 Transient vs. Steady Dynamics

The dynamics, as well as the control, perspectives are quite different depending on whether the mechanical interest in the background technical application is towards the transient or the steady system response. In the problem of ship capsizing, for example, it has been emphasised that short-term transient behaviour is particularly

important [153]. As a matter of fact, there are situations where a temporary escape from the potential well may be unessential for the system operating effectiveness, whereas it has to be strictly avoided in other practical situations. The transient regime is very important and governs the whole system performance in the non-linear dynamics due to short-term excitations (such as those associated with impact forces or seismic loads), while the steady state dynamics are apparently of major interest in the presence of stationary, long-term, excitations. Moreover, the transient regime has obviously as minor importance as shorter it is (like it occurs, e.g., in highly damped systems) while it becomes crucial when it is long (e.g., in highly deformable systems).

All of the previous *mechanical*, *bifurcational*, and *dynamical* distinguishing features have meaningful influences on the realizable control scenarios.

4.10.7 Overall vs. Localized Control

Among various possible categorization perspectives, the broad one in this subsection title basically distinguishes between overall and localized control, and seems to be a rather comprehensive, as well as comparatively useful, one. As a matter of fact, it strictly depends on the two previously addressed, distinct, aspects of the whole problem, namely (i) the *main* or *secondary* character of the bifurcational event of interest, and (ii) the *large* or *small* scale nature of the ensuing dynamics.

As already said, the global bifurcations of the manifolds of a hilltop saddle are main events organizing the system dynamics on a large scale. Nonetheless, in terms of control, a basic distinction does apply as to the possibility (or the interest) to control the global bifurcations, and the ensuing system dynamics, over the whole phase space or just one meaningful part of it. Yet, within this general framework, different situations must be considered.

1. The pursued distinction makes actually full sense only in systems exhibiting (at least) two separate homoclinic orbits, organized by one intermediate hilltop saddle and surrounding two clearly identified topological regions in the phase space, which correspond to the two potential wells: this is the case, e.g., of the Duffing-type (smooth or non-smooth) hardening oscillators. In such systems, one can aim at controlling only one of the two homoclinic bifurcations, which also entails controlling the dynamics in one clearly identified part of the phase space: for apparent reasons, this topologically localized control strategy has been termed “one-side control” in Sect. 4.9. Alternatively, one can aim at controlling both the two homoclinic bifurcations, which clearly corresponds to controlling, on average, the dynamics in the whole phase space: this topologically overall control strategy naturally has been termed “global control” in Sect. 4.9.
2. A similar distinction applies to the escape systems which exhibit two separate heteroclinic orbits connecting two organizing hilltop saddles and surrounding a unique potential well: this is the case of the Duffing-type softening oscillator or of the rocking block. Yet, herein, the distinction is considerably weaker from the application viewpoint. Indeed, also in such systems one can aim at

controlling either only one of the two heteroclinic bifurcations, thus realizing a “one-side control,” or both heteroclinic bifurcations, thus realizing a “global control.” However, the former strategy no more entails controlling the dynamics in one clearly identified part of the phase space or, in other terms, it no more produces a topologically localized control. What actually happens is that the perturbed heteroclinic loop responsible for complex dynamic phenomena (such as a chaotic saddle and the ensuing horseshoe dynamics) actually breaks down, but the heteroclinic intersection of the non-controlled orbit still allows for the penetration of tongues of the basin of attraction of the infinity into the in-well bounded basins, which may ultimately result in a much earlier system failure.

3. The third case of interest, within the large scale dynamics ensuing from one main bifurcational event, is concerned with all escaping situations characterized by only one homoclinic orbit, which delimits a clearly defined part of the phase space: this is the case of all Helmholtz-type softening oscillators [137]. For consistency with the previously adopted terminology, controlling the solely possible homoclinic bifurcation corresponds to realizing a “one-side control,” which in this case actually entails controlling the sole topologically meaningful part of the system phase space.

The second distinction between overall and localized control is concerned with the secondary bifurcational events possibly associated with local saddles, and with the ensuing system dynamics. It refers to the possibility, and the interest, to control not so much the global bifurcations of the invariant manifolds of the hilltop saddle organizing the whole dynamics as those of the manifolds of some non-hilltop saddle governing the dynamics within a restricted area of the phase space and making sense in specific technical situations. Yet, at least two schematically different situations may occur, again.

1. If the secondary event entails the onset of a localized – though possibly complex – dynamics on a *small scale* (e.g., in-well chaos), only the “one-side control” (from the large scale dynamics perspective mentioned at last point 3) can of course be applied.
2. If, on the contrary, due to the involvement of also the hilltop saddle through an heteroclinic connection, the secondary bifurcational event entails a *large scale* dynamics (think, e.g., of the transition from a single-well to a cross-well chaos in a specific hardening Duffing oscillator), one can again distinguish between different control strategies (i.e., “global” vs. “one-side”) along the lines previously discussed.

Yet, there is a fundamental difference with respect to the hilltop saddle manifolds bifurcation. Indeed, while in this case there can be a very useful chance to develop an *analytical*, either exact or approximate, treatment [63], no analytical tools are usually available to detect specific homo/heteroclinic bifurcations of other saddles playing an only locally dominant role in the dynamics. In such cases, the lack of mathematical tools requires adopting a fully *numerical* approach, wherein both the invariant manifolds and their bifurcations are detected through computer simula-

tions [136], with obvious restrictions in terms of both generality and applicability of the procedure.

4.10.8 System-Independent vs. System-Dependent Controls

With reference to the control scenario on the large scale, it is also worth mentioning some further, very important, control aspects associated with realizing a “one-side” or “global” control as well as with considering hardening symmetric or asymmetric systems.

They basically consist in the occurrence of meaningful generic (or sub-universal) features of the mathematical problems of optimization governing the cases of “one-side control” of whatever (softening or hardening) systems and of “global control” of hardening symmetric systems, which also entail very desirable system-independent solutions. This is in contrast with the non-generic optimal “global” control of hardening asymmetric systems, whose solutions are indeed system-dependent and may actually aim at different outcomes of the controlled dynamics, which are strictly associated with the existing asymmetry features.

4.10.9 Finite- vs. Infinite-Dimensional Systems

Finally, the very important issue of the finite- vs. infinite- dimensionality of the system has to be mentioned. The matter was already addressed in Sect. 1, when dwelling on the reliability of the reduced order models usually employed for analyzing the nonlinear dynamics and control of mechanical and structural systems.

In the framework of the present control method, from a technical point of view, detecting the unwanted global bifurcation may be quite difficult in infinite-dimensional systems, both analytically and numerically, especially because the manifolds are no longer 1D.

Referring to a buckled beam with variable boundary conditions, it has been shown [143] how the (minimal) reduced order model is able to provide the same homoclinic bifurcation threshold and the same optimal control as the actual infinite-dimensional system in the background, although this is accomplished in an exact way or with a practically acceptable approximation depending on some problem characteristics. However, in all cases, the correct infinite-dimensional analysis may add useful new resources for control purposes, thus enlarging the potentiality of the method. For example, in infinite-dimensional systems one can determine and use also the “optimal” *spatial* shape of the controlling excitation, in addition to the optimal *temporal* one solely obtainable in single d.o.f. systems [143].

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